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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Properties of Matter

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 235 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A liquid of density $600\ \mathrm{kg/m^3}$ flowing steadily in a tube of varying cross-section. The cross-section at a point A is $1.0\ \mathrm{cm^2}$ and that at B is $20\ \mathrm{mm^2}$. Both the points A and B are in same horizontal plane, the speed of the liquid at A is $10\ \mathrm{cm/s}$. The difference in pressures at A and B points is ____ Pa.

Open solution

(A) 18 (B) 144 (C) 36 (D) 72

Q2. JEE Main 2026 (Online) 8th April Evening Shift

A spherical liquid drop of radius R acquires the terminal velocity v_1 when falls through a gas of viscosity η. Now the drop is broken into 64 identical droplets and each droplet acquires terminal velocity v_2 falling through the same gas. The ratio of terminal velocities v_1/v_2 is ____ .

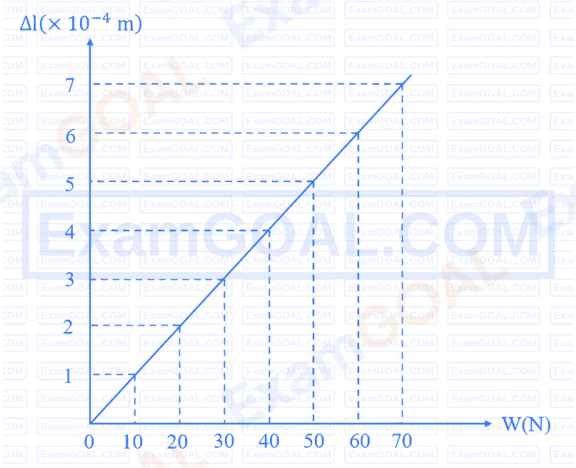
Open solution

(A) 4 (B) 0.25 (C) 32 (D) 16

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with a load W connected to the other end. If the cross-sectional area of the wire is $10^{-5}\ \mathrm{m^2}$ then the Young's modulus of the wire is ____ $\mathrm{N/m^2}$.

Open solution



(A) 1.0×10^{11} (B) 2.0×10^{10} (C) 1.0×10^{10} (D) 2.0×10^{11}

Q4. JEE Main 2026 (Online) 6th April Evening Shift

A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is $528\mathrm{dm^3}$ (dm : decimeter) The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is ____ cm .

Open solution

(A) $120\sqrt{2}$ (B) $140\sqrt{2}$ (C) $140\sqrt{3}$ (D) $120\sqrt{3}$

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Q5. JEE Main 2026 (Online) 6th April Morning Shift

A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\ \mathrm{N/m^2}$ and its radius is 4 mm , then maximum acceleration the lift can take is ____ $\mathrm{m/s^2}$.

(take $\mathrm{g} = 10\ \mathrm{m/s^2}$ and $\pi = 3.14$)

(A) 2.56 (B) 3.89 (C) 4.32 (D) 5.16

Open solution

Q6. JEE Main 2026 (Online) 6th April Morning Shift

The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m , then the length of wire B is

____ m.

(A) 1.1 (B) 2.22 (C) 1.21 (D) 4.44

Open solution

Q7. JEE Main 2026 (Online) 5th April Evening Shift

Eight mercury drops, each of radius r, coalesce to form a bigger drop. The surface energy released in this process is ____ - (S is the surface tension of mercury).

(A) $8\pi r^2\ \mathrm{S}$ (B) $16\pi r^2 S$ (C) $64\pi r^2 S$ (D) $4\pi r^2\ \mathrm{S}$

Open solution

Q8. JEE Main 2026 (Online) 5th April Morning Shift

The Young's modulus of steel wire of radius r and length L is Y.

If the radius r and length L of the wire are doubled then the value of Y

(A) increases by two times (B) reduces by half

(C) remains unchanged (D) becomes one fourth

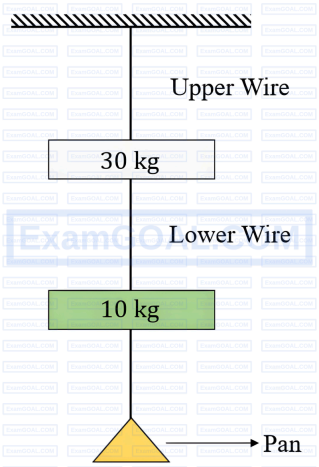
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Q9. JEE Main 2026 (Online) 5th April Morning Shift

Two wires as shown in the figure below, made of steel and have breaking stress of $12 \times 10^8\ \mathrm{N/m^2}$. Area of cross-section of upper wire is $0.008\ \mathrm{cm^2}$ and of lower wire is $0.004\ \mathrm{cm^2}$. The maximum mass that can be added to pan without breaking any wire is ____ kg.

Open solution



(A) 56 (B) 38 (C) 96 (D) 5.6

Q10. JEE Main 2026 (Online) 4th April Evening Shift

A metal string A is suspended from a rigid support and its free end is attached to a block of mass M. Second block having mass 2 M is suspended at the bottom of the first block using a string B. The area of cross sections of strings A and B are same. The ratio of lengths of strings of A to B is 2 and the ratio of their Young's moduli (Y_A/Y_B) is 0.5 . The ratio of elongations in A to B is ____ .

Open solution

(A) 1 (B) 4 (C) 8 (D) 6

Q11. JEE Main 2026 (Online) 4th April Evening Shift

A water spray gun is attached to a hose of cross sectional area $30\ \mathrm{cm^2}$. The gun comprises of 10 perforations each of cross sectional area of $15\ \mathrm{mm^2}$. If the water flows in the hose with the speed of $50\ \mathrm{cm/s}$, calculate the speed at which the water flows out from each perforation. (Neglect any edge effects)

Open solution

(A) $100\ \mathrm{m/s}$ (B) $10\ \mathrm{m/s}$
(C) $1000\ \mathrm{m/s}$ (D) $15 \times 10^2\ \mathrm{m/s}$

Q12. JEE Main 2026 (Online) 4th April Morning Shift

The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^7\ \mathrm{N/m^2}$. The percentage decrease in the volume is ____ .

(Bulk modulus of water $= 2.1 \times 10^9\ \mathrm{N/m^2}$.)

Open solution

(A) 2 % (B) 3 % (C) 6 % (D) 4 %

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Q13. JEE Main 2026 (Online) 4th April Morning Shift

A string A of length 0.314 m and Young's modulus $2 \times 10^{10}\ \mathrm{N/m^2}$ is connected to another string B of length and Young's modulus both twice of those of A. This series combination of strings is then suspended from a rigid support and its free end is fixed to a load of mass 0.8 kg . The net change in length of the combination is ____ mm.

(radius of both the strings is 0.2 mm and acceleration due to gravity $= 10\ \mathrm{m/s^2}$) (Mass of both strings is to be neglected as compared to the mass of load)

(A) 3 (B) 2 (C) 1.9 (D) 1

Open solution

Q14. JEE Main 2026 (Online) 2nd April Evening Shift

The surface tension of a soap bubble is 0.03 N/m. The work done in increasing the diameter of bubble from 2 cm to 6 cm is $\alpha\pi \times 10^{-4}$ J. The value of α is _____. (Take $\pi = 3.14$)

(A) 0.86 (B) 0.64 (C) 1.92 (D) 7.68

Open solution

Q15. JEE Main 2026 (Online) 2nd April Evening Shift

If an air bubble of diameter 2 mm rises steadily through a liquid of density 2000 kg/m^3 at a rate of 0.5 cm/s, then the coefficient of viscosity of liquid is _____ Poise. (Take g = 10 m/s^2)

(A) 0.88 (B) 8.8 (C) 88.8 (D) 0.088

Open solution

Q16. JEE Main 2026 (Online) 2nd April Morning Shift

A liquid drop of diameter 2 mm breaks into 512 droplets. The change in surface energy is $\alpha \times 10^{-6}$ J. The value of α is _____. (Take surface tension of liquid = 0.08 N/m)

(A) 10 (B) 7 (C) 8 (D) 11

Open solution

Q17. JEE Main 2026 (Online) 28th January Morning Shift

Two wires A and B made of different materials of lengths 6.0 cm and 5.4 cm , respectively and area of cross sections $3.0 \times 10^{-5}\ \mathrm{m^2}$ and $4.5 \times 10^{-5}\ \mathrm{m^2}$, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of A to that of B is $x : 3$. The value of x is ____

(A) 2 (B) 1 (C) 4 (D) 5

Open solution

Q18. JEE Main 2026 (Online) 24th January Evening Shift

A cubical block of density $\rho_b = 600\ \mathrm{kg/m^3}$ floats in a liquid of density $\rho_e = 900\mathrm{kg/m^3}$. If the height of block is $H = 8.0\ \mathrm{cm}$ then height of the submerged part is

____ cm .

(A) 6.3 (B) 4.3 (C) 7.3 (D) 5.3

Open solution

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Q19. JEE Main 2026 (Online) 24th January Morning Shift

A brass wire of length 2 m and radius 1 mm at 27°C is held taut between two rigid supports. Initially it was cooled to a temperature of -43°C creating a tension T in the wire. The temperature to which the wire has to be cooled in order to increase the tension in it to $1.4T$, is ____ $^\circ\text{C}$.

Open solution

(A) -71 (B) -80 (C) -65 (D) -86

Q20. JEE Main 2026 (Online) 23rd January Evening Shift

A small metallic sphere of diameter 2 mm and density $10.5\ \text{g/cm}^3$ is dropped in glycerine having viscosity 10 Poise and density $1.5\ \text{g/cm}^3$ respectively. The terminal velocity attained by the sphere is ____ cm/s.

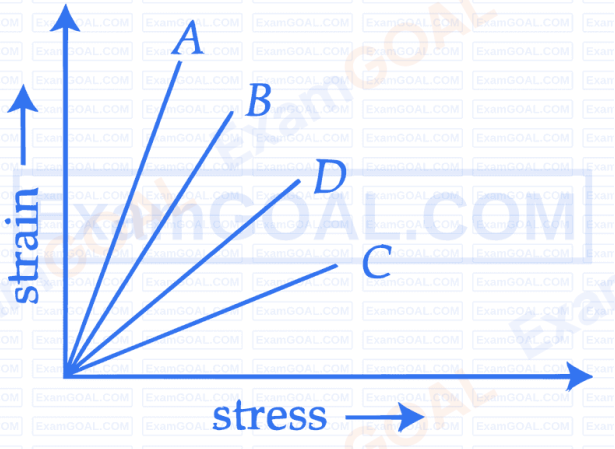
$\left(\pi = \frac{22}{7} \text{ and } g = 10\ \text{m/s}^2\right)$

Open solution

(A) 1.0 (B) 1.5 (C) 3.0 (D) 2.0

Q21. JEE Main 2026 (Online) 23rd January Morning Shift

The strain-stress plot for materials A, B, C and D is shown in the figure. Which material has the largest Young's modulus ?



Open solution

(A) B (B) A (C) D (D) C

Q22. JEE Main 2026 (Online) 22nd January Evening Shift

When a part of a straight capillary tube is placed vertically in a liquid, the liquid raises upto certain height h. If the inner radius of the capillary tube, density of the liquid and surface tension of the liquid decrease by 1% each, then the height of the liquid in the tube will change by ____ $\%$.

Open solution

(A) +3 (B) +1 (C) -1 (D) -3

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Q23. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

Statement I : Pressure of a fluid is exerted only on a solid surface in contact as the fluid-pressure does not exist everywhere in a still fluid.

Statement II : Excess potential energy of the molecules on the surface of a liquid, when compared to interior, results in surface tension.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q24. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Surface tension of two liquids (having same densities), T_1 and T_2, are measured using capillary rise method utilizing two tubes with inner radii of r_1 and r_2 where $r_1 > r_2$. The measured liquid heights in these tubes are h_1 and h_2 respectively. [Ignore the weight of the liquid above the lowest point of miniscus]. The heights h_1 and h_2 and surface tensions T_1 and T_2 satisfy the relation :

(A) $h_1 > h_2$ and $T_1 = T_2$

(B) $h_1 < h_2$ and $T_1 = T_2$

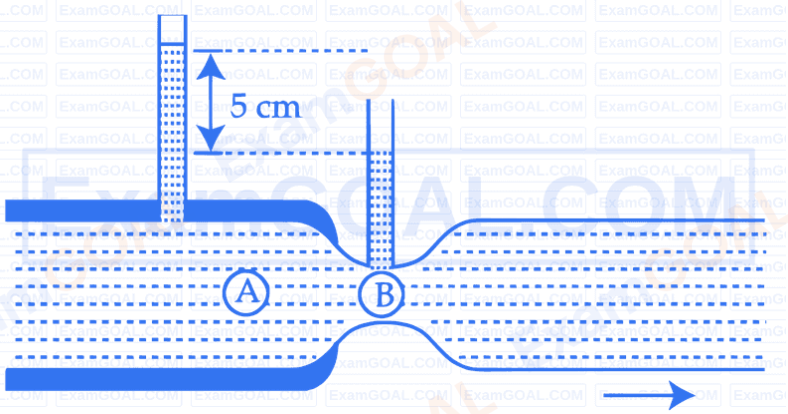
(C) $h_1 > h_2$ and $T_1 < T_2$

(D) $h_1 = h_2$ and $T_1 = T_2$

Q25. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at A and B are $6\ \text{cm}^2$ and $3\ \text{cm}^2$ respectively. The rate of flow will be ____ cm^3/s. (take $g = 10\ \text{m/s}^2$)



(A) $100\sqrt{3}$

(B) $\frac{200}{\sqrt{3}}$

(C) $200\sqrt{6}$

(D) $200\sqrt{3}$

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Q26. JEE Main 2025 (Online) 8th April Evening Shift

A 3 m long wire of radius 3 mm shows an extension of 0.1 mm when loaded vertically by a mass of 50 kg in an experiment to determine Young's modulus. The value of Young's modulus of the wire as per this experiment is $P \times 10^{11}$ Nm^{-2}, where the value of P is: (Take $g = 3\pi$ m/s^2)

Open solution

(A) 2.5 (B) 25 (C) 10 (D) 5

Q27. JEE Main 2025 (Online) 7th April Evening Shift

A capillary tube of radius 0.1 mm is partly dipped in water (surface tension 70 dyn/cm and glass water contact angle ≈ 0°) with 30° inclined with the vertical. The length of water risen in the capillary is _____ cm. (Take $g = 9.8$ m/s^2)

Open solution

(A) $\frac{82}{5}$ (B) $\frac{68}{5}$ (C) $\frac{57}{2}$ (D) $\frac{71}{5}$

Q28. JEE Main 2025 (Online) 7th April Morning Shift

Two wires A and B are made of same material having ratio of lengths $\frac{L_A}{L_B} = \frac{1}{3}$ and their diameters ratio $\frac{d_A}{d_B} = 2$. If both the wires are stretched using same force, what would be the ratio of their respective elongations?

Open solution

(A) $3:4$ (B) $1:12$ (C) $1:3$ (D) $1:6$

Q29. JEE Main 2025 (Online) 4th April Evening Shift

A cylindrical rod of length 1 m and radius 4 cm is mounted vertically. It is subjected to a shear force of 10^5 N at the top. Considering infinitesimally small displacement in the upper edge, the angular displacement θ of the rod axis from its original position would be : (shear moduli, $G = 10^{10}$ N/m^2)

Open solution

(A) $1/160\pi$ (B) $1/2\pi$ (C) $1/4\pi$ (D) $1/40\pi$

Q30. JEE Main 2025 (Online) 4th April Morning Shift

Two liquids A and B have θ_A and θ_B as contact angles in a capillary tube. If $K = \cos\theta_A / \cos\theta_B$, then identify the correct statement:

Open solution

(A) K is negative, then liquid A and liquid B have convex meniscus.

(B) K is negative, then liquid A and liquid B have concave meniscus.

(C) K is zero, then liquid A has convex meniscus and liquid B has concave meniscus.

(D) K is negative, then liquid A has concave meniscus and liquid B has convex meniscus.

Q31. JEE Main 2025 (Online) 3rd April Evening Shift

Two cylindrical vessels of equal cross sectional area of $2\ \text{m}^2$ contain water upto heights 10 m and 6 m , respectively. If the vessels are connected at their bottom then the work done by the force of gravity is (Density of water is $10^3\ \text{kg/m}^3$ and $\text{g} = 10\ \text{m/s}^2$)

Open solution

(A) 1×10^5 J (B) 4×10^4 J (C) 8×10^4 J (D) 6×10^4 J

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Q32. JEE Main 2025 (Online) 3rd April Evening Shift

A solid steel ball of diameter 3.6 mm acquired terminal velocity $2.45 \times 10^{-2}\ \mathrm{m/s}$ while falling under gravity through an oil of density $925\ \mathrm{kg\ m^{-3}}$. Take density of steel as $7825\ \mathrm{kg\ m^{-3}}$ and g as $9.8\ \mathrm{m/s^2}$. The viscosity of the oil in SI unit is

Open solution

(A) 2.18 (B) 1.68 (C) 2.38 (D) 1.99

Q33. JEE Main 2025 (Online) 3rd April Morning Shift

Consider a completely full cylindrical water tank of height 1.6 m and of cross-sectional area $0.5\ \mathrm{m^2}$. It has a small hole in its side at a height 90 cm from the bottom. Assume, the cross-sectional area of the hole to be negligibly small as compared to that of the water tank. If a load 50 kg is applied at the top surface of the water in the tank then the velocity of the water coming out at the instant when the hole is opened is:

$(\mathrm{g} = 10\ \mathrm{m/s^2})$

Open solution

(A) $2\ \mathrm{m/s}$ (B) $5\ \mathrm{m/s}$ (C) $3\ \mathrm{m/s}$ (D) $4\ \mathrm{m/s}$

Q34. JEE Main 2025 (Online) 2nd April Evening Shift

Two water drops each of radius ' r ' coalesce to form a bigger drop. If ' T ' is the surface tension, the surface energy released in this process is :

Open solution

(A) $4\pi \mathrm{r}^2\ \mathrm{T}\left[2 - 2^{1/3}\right]$ (B) $4\pi \mathrm{r}^2\ \mathrm{T}[1 + \sqrt{2}]$

(C) $4\pi \mathrm{r}^2\ \mathrm{T}\left[2 - 2^{2/3}\right]$ (D) $4\pi \mathrm{r}^2\ \mathrm{T}[\sqrt{2} - 1]$

Q35. JEE Main 2025 (Online) 29th January Morning Shift

The fractional compression $\left(\frac{\Delta V}{V}\right)$ of water at the depth of 2.5 km below the sea level is _____ %. Given, the Bulk modulus of water = 2×10^9 N m^{-2}, density of water = 10^3 kg m^{-3}, acceleration due to gravity $g = 10$ m s^{-2}.

Open solution

(A) 1.0 (B) 1.25 (C) 1.75 (D) 1.5

Q36. JEE Main 2025 (Online) 28th January Evening Shift

A 400 g solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water?

(Given: density of water = 1000 kg m^{-3})

Open solution

(A) 400 cm^3 (B) 600 cm^3 (C) 1400 cm^3 (D) 4000 cm^3

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Q37. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

In the experiment for measurement of viscosity ' η ' of given liquid with a ball having radius R, consider following statements.

A. Graph between terminal velocity V and R will be a parabola.

B. The terminal velocities of different diameter balls are constant for a given liquid.

C. Measurement of terminal velocity is dependent on the temperature.

D. This experiment can be utilized to assess the density of a given liquid.

E. If balls are dropped with some initial speed, the value of η will change.

Choose the correct answer from the options given below:

(A) C, D and E Only (B) A, B and E Only (C) A, C and D Only (D) B, D and E Only

Q38. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Consider following statements:

A. Surface tension arises due to extra energy of the molecules at the interior as compared to the molecules at the surface, of a liquid.

B. As the temperature of liquid rises, the coefficient of viscosity increases.

C. As the temperature of gas increases, the coefficient of viscosity increases

D. The onset of turbulence is determined by Reynold's number.

E. In a steady flow two stream lines never intersect.

Choose the correct answer from the options given below:

(A) B, C, D Only (B) C, D, E Only (C) A, D, E Only (D) A, B, C Only

Q39. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An air bubble of radius 0.1 cm lies at a depth of 20 cm below the free surface of a liquid of density $1000\ \mathrm{kg/m^3}$. If the pressure inside the bubble is $2100\ \mathrm{N/m^2}$ greater than the atmospheric pressure, then the surface tension of the liquid in SI unit is (use $g = 10\ \mathrm{m/s^2}$)

(A) 0.02 (B) 0.05 (C) 0.25 (D) 0.1

Q40. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

The amount of work done to break a big water drop of radius ' R ' into 27 small drops of equal radius is 10 J . The work done required to break the same big drop into 64 small drops of equal radius will be

(A) 20 J (B) 10 J (C) 5 J (D) 15 J

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Q41. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Water flows in a horizontal pipe whose one end is closed with a valve. The reading of the pressure gauge attached to the pipe is P_1. The reading of the pressure gauge falls to P_2 when the valve is opened. The speed of water flowing in the pipe is proportional to

(A) $(P_1 - P_2)^2$ (B) $\sqrt{P_1 - P_2}$ (C) $P_1 - P_2$ (D) $(P_1 - P_2)^4$

Q42. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Given below are two statements:

Statement I: The hot water flows faster than cold water

Statement II: Soap water has higher surface tension as compared to fresh water. In the light above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

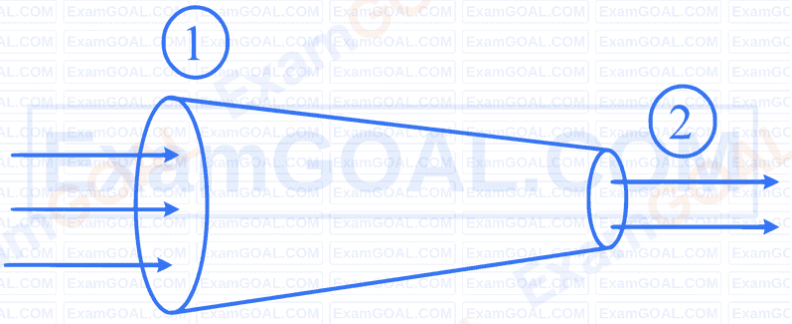
(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q43. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution



A tube of length L is shown in the figure. The radius of cross section at the point (1) is 2 cm and at the point (2) is 1 cm , respectively. If the velocity of water entering at point (1) is $2\,\mathrm{m/s}$, then velocity of water leaving the point (2) will be

(A) 4 m/s (B) 6 m/s (C) 2 m/s (D) 8 m/s

Q44. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

A small rigid spherical ball of mass M is dropped in a long vertical tube containing glycerine. The velocity of the ball becomes constant after some time. If the density of glycerine is half of the density of the ball, then the viscous force acting on the ball will be (consider g as acceleration due to gravity)

(A) 2 Mg (B) $\frac{3}{2}\mathrm{Mg}$ (C) $\frac{\mathrm{Mg}}{2}$ (D) Mg

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Q45. JEE Main 2024 (Online) 9th April Evening Shift

The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is:

(A) $1:9$ (B) $1:27$ (C) $1:81$ (D) $1:3$

Open solution

Q46. JEE Main 2024 (Online) 9th April Evening Shift

A spherical ball of radius $1 \times 10^{-4}\ \mathrm{m}$ and density $10^5\ \mathrm{kg/m^3}$ falls freely under gravity through a distance h before entering a tank of water. If after entering in water the velocity of the ball does not change, then the value of h is approximately:

(The coefficient of viscosity of water is $9.8 \times 10^{-6}\ \mathrm{N\,s/m^2}$)

(A) 2518 m (B) 2396 m (C) 2249 m (D) 2296 m

Open solution

Q47. JEE Main 2024 (Online) 9th April Morning Shift

A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is :

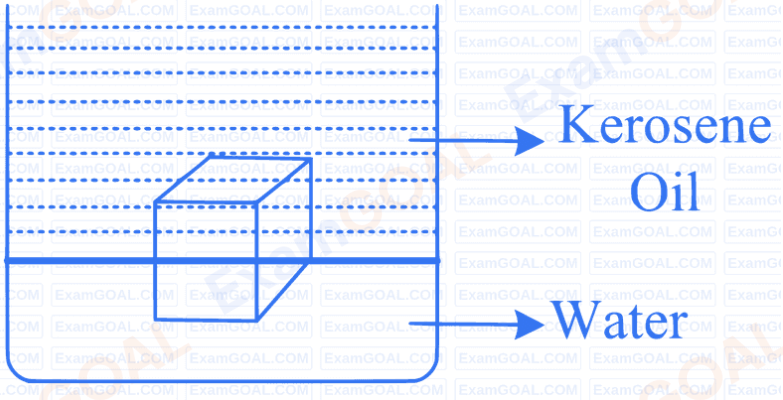
(A) $\left(\frac{\sigma-2}{\sigma+2}\right)^{1/3}$ (B) $\left(\frac{\sigma+1}{\sigma-1}\right)^{1/3}$

(C) $\left(\frac{\sigma-1}{\sigma}\right)^{1/3}$ (D) $\left(\frac{\sigma}{\sigma-1}\right)^{1/3}$

Open solution

Q48. JEE Main 2024 (Online) 8th April Evening Shift

A cube of ice floats partly in water and partly in kerosene oil. The ratio of volume of ice immersed in water to that in kerosene oil (specific gravity of Kerosene oil = 0.8, specific gravity of ice = 0.9):



(A) 5 : 4 (B) 9 : 10 (C) 8 : 9 (D) 1 : 1

Open solution

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Q49. JEE Main 2024 (Online) 8th April Morning Shift

Young's modulus is determined by the equation given by $\mathrm{Y} = 49000 \frac{\mathrm{m}}{\mathrm{l}} \frac{\mathrm{dyne}}{\mathrm{cm}^2}$ where M is the mass and l is the extension of wire used in the experiment. Now error in Young modules (Y) is estimated by taking data from $M - l$ plot in graph paper. The smallest scale divisions are $5\ \mathrm{g}$ and $0.02\ \mathrm{cm}$ along load axis and extension axis respectively. If the value of M and l are $500\ \mathrm{g}$ and $2\ \mathrm{cm}$ respectively then percentage error of Y is :

Open solution

(A) 2% (B) 0.02% (C) 0.5% (D) 0.2%

Q50. JEE Main 2024 (Online) 8th April Morning Shift

Correct Bernoulli's equation is (symbols have their usual meaning) :

Open solution

(A) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$ (B) $P + mgh + \frac{1}{2}mv^2 = \text{constant}$

(C) $P + \rho gh + \rho v^2 = \text{constant}$ (D) $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Q51. JEE Main 2024 (Online) 6th April Evening Shift

Pressure inside a soap bubble is greater than the pressure outside by an amount : (given : $\mathrm{R} =$ Radius of bubble, $\mathrm{S} =$ Surface tension of bubble)

Open solution

(A) $\frac{S}{R}$ (B) $\frac{4\,\mathrm{S}}{\mathrm{R}}$ (C) $\frac{4\mathrm{R}}{\mathrm{S}}$ (D) $\frac{2S}{R}$

Q52. JEE Main 2024 (Online) 6th April Morning Shift

A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0. After sometime, the ball falls with constant velocity. The viscous force on the ball is :

Open solution

(A) $mg\left(1 - \frac{\rho_0}{\rho}\right)$ (B) $mg\left(\frac{\rho_0}{\rho} - 1\right)$

(C) $mg\left(1 - \rho\rho_0\right)$ (D) $mg\left(1 + \frac{\rho}{\rho_0}\right)$

Q53. JEE Main 2024 (Online) 5th April Evening Shift

Match List I with List II :

Open solution

	LIST I		LIST II
A.	A force that restores an elastic body of unit area to its original state	I.	Bulk modulus
B.	Two equal and opposite forces parallel to opposite faces	II.	Young's modulus
C.	Forces perpendicular everywhere to the surface per unit area same everywhere	III.	Stress
D.	Two equal and opposite forces perpendicular to opposite faces	IV.	Shear modulus

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) (B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(II), (B)-(IV), (C)-(I), (D)-(III) (D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

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Q54. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements :

Statement I : When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary. The contact angle may be 0°.

Statement II : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

In the light of the above statement, choose the correct answer from the options given below.

(A) Statement I is true and Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q55. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Given below are two statements :

Statement I : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

Statement II : The rise of a liquid in a capillary tube does not depend on the inner radius of the tube.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false.

(B) Both Statement I and Statement II are true.

(C) Statement I is false but Statement II is true.

(D) Statement I is true but Statement II is false.

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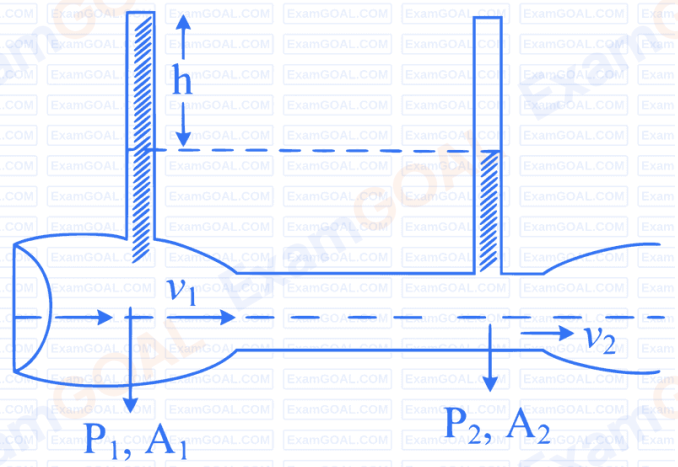
Q56. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Given below are two statements :

Statement I : When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $\mathrm{P}_1 - \mathrm{P}_2 = \rho g\,(\,\mathrm{h}_2 - \mathrm{h}_1)$.

Statement II : In ventury tube shown $2\mathrm{gh} = v_1^2 - v_2^2$



In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is correct but Statement II is incorrect.

(B) Both Statement I and Statement II are correct.

(C) Both Statement I and Statement II are incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q57. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A big drop is formed by coalescing 1000 small droplets of water. The surface energy will become :

(A) $\frac{1}{100}$ th (B) $\frac{1}{10}$ th (C) 100 times (D) 10 times

Q58. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

With rise in temperature, the Young's modulus of elasticity :

(A) changes erratically (B) increases

(C) decreases (D) remains unchanged

Q59. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

A small spherical ball of radius r, falling through a viscous medium of negligible density has terminal velocity 'v'. Another ball of the same mass but of radius $2r$, falling through the same viscous medium will have terminal velocity:

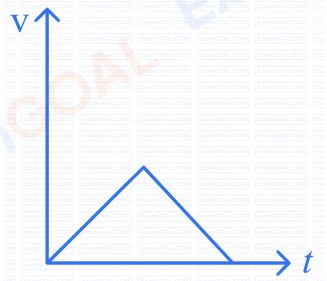
(A) $4\mathrm{v}$ (B) $2\mathrm{v}$ (C) $\frac{v}{4}$ (D) $\frac{\mathrm{v}}{2}$

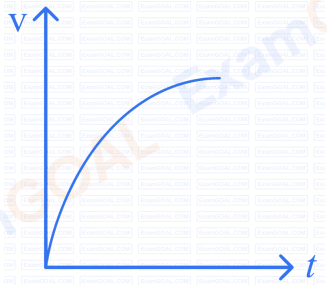
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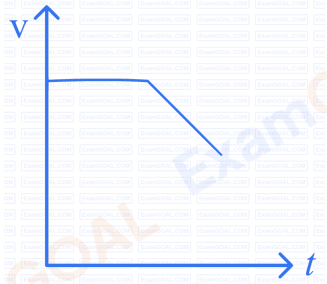
Q60. JEE Main 2024 (Online) 31st January Morning Shift

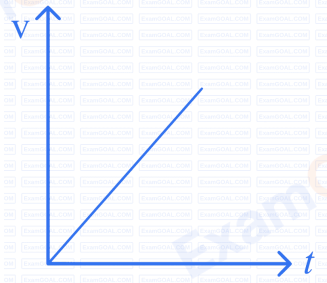
A small steel ball is dropped into a long cylinder containing glycerine. Which one of the following is the correct representation of the velocity time graph for the transit of the ball?

Open solution

(A) 

(B) 

(C) 

(D) 

Q61. JEE Main 2024 (Online) 30th January Morning Shift

Young's modulus of material of a wire of length 'L' and cross-sectional area A is Y. If the length of the wire is doubled and cross-sectional area is halved then Young's modulus will be :

(A) 4 Y (B) 2 Y (C) $\frac{Y}{4}$ (D) Y

Open solution

Q62. JEE Main 2024 (Online) 29th January Evening Shift

A small liquid drop of radius R is divided into 27 identical liquid drops. If the surface tension is T, then the work done in the process will be:

(A) $4\pi R^2\ T$ (B) $8\pi R^2\ T$ (C) $\frac{1}{8}\pi R^2 T$ (D) $3\pi R^2\ T$

Open solution

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Q63. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F, its length increases by l. If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become:

(A) 2 times **(B)** 4 times **(C)** 3 times **(D)** $\frac{3}{2}$ times

Q64. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Given below are two statements:

Statement I : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in hot water.

Statement II : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in cold water.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q65. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) Both (A) and (R) are true and (R) is the correct explanation of (A)

(D) (A) is true but (R) is false

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Q66. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Given below are two statements :

Statement (I) :Viscosity of gases is greater than that of liquids.

Statement (II) : Surface tension of a liquid decreases due to the presence of insoluble impurities.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

Q67. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

A wire of length ' L ' and radius ' r ' is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by ' l '. Another wire of same material of length ' $2\,\mathrm{L}$ ' and radius ' $2r$ ' is pulled by a force ' $2f$ '. Then the increase in its length will be :

(A) $2l$ (B) $4l$ (C) l (D) $l/2$

Q68. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : A spherical body of radius $(5 \pm 0.1)\mathrm{mm}$ having a particular density is falling through a liquid of constant density. The percentage error in the calculation of its terminal velocity is 4%.

Reason R : The terminal velocity of the spherical body falling through the liquid is inversely proportional to its radius.

In the light of the above statements, choose the correct answer from the options given below

(A) A is false but $\mathbf{R}$ is true

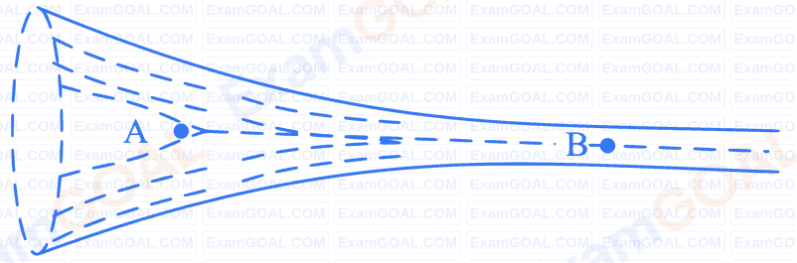
(B) A is true but $\mathbf{R}$ is false

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

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Q69. JEE Main 2023 (Online) 13th April Morning Shift



Open solution

The figure shows a liquid of given density flowing steadily in horizontal tube of varying cross - section. Cross sectional areas at A is $1.5\ \mathrm{cm}^2$, and B is $25\ \mathrm{mm}^2$, if the speed of liquid at B is $60\ \mathrm{cm/s}$ then $(\mathrm{P_A} - \mathrm{P_B})$ is :

(Given $\mathrm{P_A}$ and $\mathrm{P_B}$ are liquid pressures at A and B points.

density $\rho = 1000\ \mathrm{kg\,m^{-3}}$

A and B are on the axis of tube

(A) $27\ \mathrm{Pa}$ (B) $175\ \mathrm{Pa}$ (C) $135\ \mathrm{Pa}$ (D) $36\ \mathrm{Pa}$

Q70. JEE Main 2023 (Online) 13th April Morning Shift

Under isothermal condition, the pressure of a gas is given by $\mathrm{P} = a\,\mathrm{V}^{-3}$, where a is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to

(A) $\frac{P}{2}$ (B) 2 P (C) 3 P (D) P

Open solution

Q71. JEE Main 2023 (Online) 11th April Evening Shift

Eight equal drops of water are falling through air with a steady speed of $10\ \mathrm{cm/s}$. If the drops coalesce, the new velocity is:-

(A) $40\ \mathrm{cm/s}$ (B) $16\ \mathrm{cm/s}$ (C) $10\ \mathrm{cm/s}$ (D) $5\ \mathrm{cm/s}$

Open solution

Q72. JEE Main 2023 (Online) 10th April Evening Shift

Young's moduli of the material of wires A and B are in the ratio of $1:4$, while its area of cross sections are in the ratio of $1:3$. If the same amount of load is applied to both the wires, the amount of elongation produced in the wires A and B will be in the ratio of

[Assume length of wires A and B are same]

(A) 1 : 12 (B) 1 : 36 (C) 12 : 1 (D) 36 : 1

Open solution

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Q73. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Given below are two statements:

Statement I : Pressure in a reservoir of water is same at all points at the same level of water.

Statement II : The pressure applied to enclosed water is transmitted in all directions equally.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q74. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

A hydraulic automobile lift is designed to lift vehicles of mass $5000\ \mathrm{kg}$. The area of cross section of the cylinder carrying the load is $250\ \mathrm{cm}^2$. The maximum pressure the smaller piston would have to bear is $\left[\text{Assume } g = 10\ \mathrm{m/s^2}\right]$

(A) $20 \times 10^{+6}\ \mathrm{Pa}$ (B) $200 \times 10^{+6}\ \mathrm{Pa}$ (C) $2 \times 10^{+5}\ \mathrm{Pa}$ (D) $2 \times 10^{+6}\ \mathrm{Pa}$

Q75. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

An air bubble of volume $1\ \mathrm{cm}^3$ rises from the bottom of a lake $40\ \mathrm{m}$ deep to the surface at a temperature of $12^{\circ}\mathrm{C}$. The atmospheric pressure is $1 \times 10^5\ \mathrm{Pa}$ the density of water is $1000\ \mathrm{kg/m^3}$ and $\mathrm{g} = 10\ \mathrm{m/s^2}$. There is no difference of the temperature of water at the depth of $40\ \mathrm{m}$ and on the surface. The volume of air bubble when it reaches the surface will be:

(A) $4\ \mathrm{cm}^3$ (B) $3\ \mathrm{cm}^3$ (C) $2\ \mathrm{cm}^3$ (D) $5\ \mathrm{cm}^3$

Q76. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

An aluminium rod with Young's modulus $Y = 7.0 \times 10^{10}\ \mathrm{N/m^2}$ undergoes elastic strain of 0.04%. The energy per unit volume stored in the rod in SI unit is:

(A) 5600 (B) 2800 (C) 11200 (D) 8400

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Q77. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: When you squeeze one end of a tube to get toothpaste out from the other end, Pascal's principle is observed.

Reason R: A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) A is not correct but R is correct

(C) A is correct but R is not correct

(D) Both A and B are correct and R is the correct explanation of A

Q78. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

A small ball of mass M and density ρ is dropped in a viscous liquid of density ρ_0. After some time, the ball falls with a constant velocity. What is the viscous force on the ball ?

(A) $\mathrm{F}=\mathrm{Mg}\left(1-\frac{\rho_{\mathrm{O}}}{\rho}\right)$

(B) $\mathrm{F}=\mathrm{Mg}\left(1+\frac{\rho}{P_o}\right)$

(C) $\mathrm{F}=\mathrm{Mg}\left(1+\frac{\rho_o}{\rho}\right)$

(D) $F=Mg\left(1 \pm \rho \rho_0\right)$

Q79. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

The Young's modulus of a steel wire of length $6\ \mathrm{m}$ and cross-sectional area $3\ \mathrm{mm}^2$, is $2 \times 10^{11}\ \mathrm{N/m}^2$. The wire is suspended from its support on a given planet. A block of mass $4\ \mathrm{kg}$ is attached to the free end of the wire. The acceleration due to gravity on the planet is $\frac{1}{4}$ of its value on the earth. The elongation of wire is (Take g on the earth $=10\ \mathrm{m/s}^2$) :

(A) 0.1 cm (B) 1 cm (C) 0.1 mm (D) 1 mm

Q80. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

A mercury drop of radius $10^{-3}\ \mathrm{m}$ is broken into 125 equal size droplets. Surface tension of mercury is $0.45\ \mathrm{Nm}^{-1}$. The gain in surface energy is :

(A) $28 \times 10^{-5}\ \mathrm{J}$ (B) $17.5 \times 10^{-5}\ \mathrm{J}$ (C) $5 \times 10^{-5}\ \mathrm{J}$ (D) $2.26 \times 10^{-5}\ \mathrm{J}$

Q81. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

For a solid rod, the Young's modulus of elasticity is $3.2 \times 10^{11} \mathrm{Nm}^{-2}$ and density is $8 \times 10^3\ \mathrm{kg\,m}^{-3}$. The velocity of longitudinal wave in the rod will be.

(A) $3.65 \times 10^3\ \mathrm{ms}^{-1}$

(B) $6.32 \times 10^3\ \mathrm{ms}^{-1}$

(C) $18.96 \times 10^3\ \mathrm{ms}^{-1}$

(D) $145.75 \times 10^3\ \mathrm{ms}^{-1}$

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Q82. JEE Main 2023 (Online) 31st January Evening Shift

Under the same load, wire A having length $5.0\ \mathrm{m}$ and cross section $2.5 \times 10^{-5}\ \mathrm{m}^2$ stretches uniformly by the same amount as another wire B of length $6.0\ \mathrm{m}$ and a cross section of $3.0 \times 10^{-5}\ \mathrm{m}^2$ stretches. The ratio of the Young's modulus of wire A to that of wire B will be :

Open solution

(A) $1:2$ (B) $1:4$ (C) $1:1$ (D) $1:10$

Q83. JEE Main 2023 (Online) 31st January Morning Shift

If 1000 droplets of water of surface tension $0.07\ \mathrm{N/m}$, having same radius $1\ \mathrm{mm}$ each, combine to form a single drop. In the process the released surface energy is :-

$\left(\mathrm{Take}\ \pi = \frac{22}{7}\right)$

Open solution

(A) $7.92 \times 10^{-4}\ \mathrm{J}$ (B) $7.92 \times 10^{-6}\ \mathrm{J}$ (C) $8.8 \times 10^{-5}\ \mathrm{J}$ (D) $9.68 \times 10^{-4}\ \mathrm{J}$

Q84. JEE Main 2023 (Online) 30th January Evening Shift

A force is applied to a steel wire 'A', rigidly clamped at one end. As a result elongation in the wire is $0.2\ \mathrm{mm}$. If same force is applied to another steel wire ' B ' of double the length and a diameter 2.4 times that of the wire ' A ', the elongation in the wire ' B ' will be (wires having uniform circular cross sections)

Open solution

(A) $6.9 \times 10^{-2}\ \mathrm{mm}$ (B) $6.06 \times 10^{-2}\ \mathrm{mm}$ (C) $2.77 \times 10^{-2}\ \mathrm{mm}$ (D) $3.0 \times 10^{-2}\ \mathrm{mm}$

Q85. JEE Main 2023 (Online) 30th January Morning Shift

The height of liquid column raised in a capillary tube of certain radius when dipped in liquid A vertically is, $5\ \mathrm{cm}$. If the tube is dipped in a similar manner in another liquid B of surface tension and density double the values of liquid A, the height of liquid column raised in liquid B would be _____ m.

Open solution

(A) 0.05 (B) 0.20 (C) 0.5 (D) 0.10

Q86. JEE Main 2023 (Online) 30th January Morning Shift

Choose the correct relationship between Poisson ratio (σ), bulk modulus (K) and modulus of rigidity (η) of a given solid object :

Open solution

(A) $\sigma = \frac{3K+2\eta}{6K+2\eta}$ (B) $\sigma = \frac{3K-2\eta}{6K+2\eta}$

(C) $\sigma = \frac{6K+2\eta}{3K-2\eta}$ (D) $\sigma = \frac{6K-2\eta}{3K-2\eta}$

Q87. JEE Main 2023 (Online) 29th January Evening Shift

A fully loaded boeing aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2. It is in level flight with a speed of 1080 km/h. If the density of air ρ is 1.2 kg m^{-3}, the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be. $(\mathrm{g} = 10\ \mathrm{m/s^2})$

Open solution

(A) 16 (B) 8 (C) 6 (D) 10

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Q88. JEE Main 2023 (Online) 29th January Morning Shift

Surface tension of a soap bubble is $2.0 \times 10^{-2} \mathrm{Nm}^{-1}$. Work done to increase the radius of soap bubble from $3.5\ \mathrm{cm}$ to $7\ \mathrm{cm}$ will be:

Take $\left[\pi = \frac{22}{7}\right]$

(A) $18.48 \times 10^{-4}\ \mathrm{J}$ (B) $5.76 \times 10^{-4}\ \mathrm{J}$ (C) $0.72 \times 10^{-4}\ \mathrm{J}$ (D) $9.24 \times 10^{-4}\ \mathrm{J}$

Open solution

Q89. JEE Main 2023 (Online) 24th January Evening Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Steel is used in the construction of buildings and bridges.

Reason R : Steel is more elastic and its elastic limit is high.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct and R is the correct explanation of A

(B) A is correct but R is not correct

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) A is not correct but R is correct

Open solution

Q90. JEE Main 2023 (Online) 24th January Evening Shift

The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as $\nu = r^a \rho^b s^c$. The values of a, b and c respectively are

(A) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (B) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

(C) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Open solution

Q91. JEE Main 2023 (Online) 24th January Morning Shift

A 100 m long wire having cross-sectional area $6.25 \times 10^{-4}\ \mathrm{m}^2$ and Young's modulus is $10^{10}\ \mathrm{Nm}^{-2}$ is subjected to a load of 250 N, then the elongation in the wire will be :

(A) $6.25 \times 10^{-6}\ \mathrm{m}$ (B) $4 \times 10^{-3}\ \mathrm{m}$ (C) $4 \times 10^{-4}\ \mathrm{m}$ (D) $6.25 \times 10^{-3}\ \mathrm{m}$

Open solution

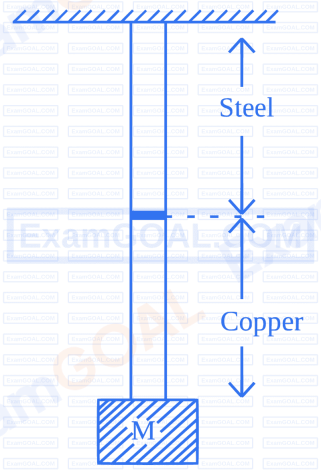
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Q92. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

A block of mass M is attached to a wire. The upper part of wire is made of steel and lower part of wire of copper (as shown in figure). Both parts have the same area of cross section. Neglecting the mass of wires, the ratio of longitudinal strain developed in steel and copper wires is :

(Given, Young's modulus of steel $= 2 \times 10^{11}\ \mathrm{N/m^2}$ and Young's modulus of copper $= 1.5 \times 10^{11}\ \mathrm{N/m^2}$)



(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$

Q93. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R): Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q94. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

(A) remain same

(B) become 8 times its initial value

(C) become $\frac{1}{4}$ of its initial value

(D) become 4 times its initial value

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Q95. JEE Main 2022 (Online) 28th July Evening Shift

A pressure-pump has a horizontal tube of cross sectional area $10\ \mathrm{cm}^2$ for the outflow of water at a speed of $20\ \mathrm{m/s}$. The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is :

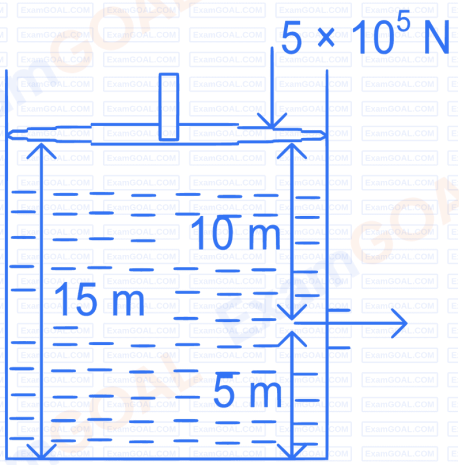
[given: density of water $= 1000\ \mathrm{kg/m^3}$]

(A) 300 N (B) 500 N (C) 250 N (D) 400 N

Open solution

Q96. JEE Main 2022 (Online) 28th July Evening Shift

Consider a cylindrical tank of radius $1\ \mathrm{m}$ is filled with water. The top surface of water is at $15\ \mathrm{m}$ from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of $5\ \mathrm{m}$ from the bottom. A force of $5 \times 10^5\ \mathrm{N}$ is applied on the top surface of water using a piston. The speed of efflux from the hole will be : (given atmospheric pressure $\mathrm{P_A} = 1.01 \times 10^5\ \mathrm{Pa}$, density of water $\rho_{\mathrm{W}} = 1000\ \mathrm{kg/m^3}$ and gravitational acceleration $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 11.6 m/s (B) 10.8 m/s (C) 17.8 m/s (D) 14.4 m/s

Open solution

Q97. JEE Main 2022 (Online) 28th July Morning Shift

A balloon has mass of $10\ \mathrm{g}$ in air. The air escapes from the balloon at a uniform rate with velocity $4.5\ \mathrm{cm/s}$. If the balloon shrinks in $5\ \mathrm{s}$ completely. Then, the average force acting on that balloon will be (in dyne).

(A) 3 (B) 9 (C) 12 (D) 18

Open solution

Q98. JEE Main 2022 (Online) 28th July Morning Shift

The force required to stretch a wire of cross-section $1\ \mathrm{cm}^2$ to double its length will be : (Given Young's modulus of the wire $= 2 \times 10^{11}\ \mathrm{N/m^2}$)

(A) $1 \times 10^7\ \mathrm{N}$ (B) $1.5 \times 10^7\ \mathrm{N}$ (C) $2 \times 10^7\ \mathrm{N}$ (D) $2.5 \times 10^7\ \mathrm{N}$

Open solution

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Q99. JEE Main 2022 (Online) 27th July Evening Shift

A steel wire of length $3.2\,\mathrm{m}$ $\left(\mathrm{Y_s} = 2.0 \times 10^{11}\,\mathrm{Nm^{-2}}\right)$ and a copper wire of length $4.4\,\mathrm{m}$ $\left(\mathrm{Y_c} = 1.1 \times 10^{11}\,\mathrm{Nm^{-2}}\right)$, both of radius $1.4\,\mathrm{mm}$ are connected end to end. When stretched by a load, the net elongation is found to be $1.4\,\mathrm{mm}$. The load applied, in Newton, will be: (Given $\pi = \frac{22}{7}$)

Open solution

(A) 360 (B) 180 (C) 1080 (D) 154

Q100. JEE Main 2022 (Online) 27th July Morning Shift

Two cylindrical vessels of equal cross-sectional area $16\,\mathrm{cm^2}$ contain water upto heights $100\,\mathrm{cm}$ and $150\,\mathrm{cm}$ respectively. The vessels are interconnected so that the water levels in them become equal. The work done by the force of gravity during the process, is [Take, density of water $= 10^3\,\mathrm{kg/m^3}$ and $\mathrm{g} = 10\,\mathrm{ms^{-2}}$] :

Open solution

(A) 0.25 J (B) 1 J (C) 8 J (D) 12 J

Q101. JEE Main 2022 (Online) 26th July Evening Shift

The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4}\,\mathrm{m^2}$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be :

Open solution

(take $g = 10\,ms^{-2}$)

(A) $6.25 \times 10^{-4}\,\mathrm{m^2}$ (B) $10 \times 10^{-4}\,\mathrm{m^2}$

(C) $1 \times 10^{-4}\,\mathrm{m^2}$ (D) $1.67 \times 10^{-4}\,\mathrm{m^2}$

Q102. JEE Main 2022 (Online) 26th July Morning Shift

A water drop of radius $1\,\mathrm{cm}$ is broken into 729 equal droplets. If surface tension of water is 75 dyne/cm, then the gain in surface energy upto first decimal place will be :

Open solution

(Given $\pi = 3.14$)

(A) $8.5 \times 10^{-4}\,\mathrm{J}$ (B) $8.2 \times 10^{-4}\,\mathrm{J}$ (C) $7.5 \times 10^{-4}\,\mathrm{J}$ (D) $5.3 \times 10^{-4}\,\mathrm{J}$

Q103. JEE Main 2022 (Online) 25th July Evening Shift

A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension 7.5×10^{-4} Ncm^{-1}. The radius of drop in cm will be :

Open solution

(g = 10 ms^{-2})

(A) $\frac{15}{\sqrt{(2\rho-\sigma)}}$ (B) $\frac{15}{\sqrt{(\rho-\sigma)}}$

(C) $\frac{3}{2\sqrt{(\rho-\sigma)}}$ (D) $\frac{3}{20\sqrt{(2\rho-\sigma)}}$

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Q104. JEE Main 2022 (Online) 30th June Morning Shift

An air bubble of negligible weight having radius r rises steadily through a solution of density σ at speed v. The coefficient of viscosity of the solution is given by :

(A) $\eta = \frac{4r\sigma g}{9v}$ (B) $\eta = \frac{2r^2\sigma g}{9v}$

(C) $\eta = \frac{2\pi r^2\sigma g}{9v}$ (D) $\eta = \frac{2r^2\sigma g}{3\pi v}$

Open solution

Q105. JEE Main 2022 (Online) 29th June Morning Shift

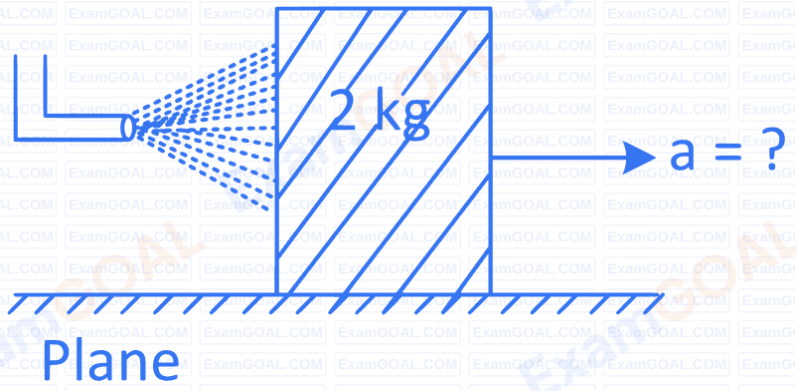
A wire of length L is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1 kg and 2 kg are suspended respectively from its free end. Then the value of L is equal to :

(A) $\sqrt{L_1L_2}$ (B) $\frac{L_1+L_2}{2}$ (C) $2L_1 - L_2$ (D) $3L_1 - 2L_2$

Open solution

Q106. JEE Main 2022 (Online) 29th June Morning Shift

A block of metal weighing 2 kg is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of 1 kgs^{-1} and at a speed of 10 ms^{-1}. Then, the initial acceleration of the block, in ms^{-2}, will be :



(A) 3 (B) 6 (C) 5 (D) 4

Open solution

Q107. JEE Main 2022 (Online) 28th June Evening Shift

A water drop of radius 1 μm falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is 1.8×10^{-5} Nsm^{-2} and its density is negligible as compared to that of water 10^6 gm^{-3}. Terminal velocity of the water drop is :

(Take acceleration due to gravity = 10 ms^{-2})

(A) 145.4×10^{-6} ms^{-1} (B) 118.0×10^{-6} ms^{-1}

(C) 132.6×10^{-6} ms^{-1} (D) 123.4×10^{-6} ms^{-1}

Open solution

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Q108. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coefficient of viscosity.

Reason R : Coefficient of viscosity = $\frac{Force}{Velocity\ gradient}$

Choose the correct answer from the options given below :

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q109. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

A water drop of diameter 2 cm is broken into 64 equal droplets. The surface tension of water is 0.075 N/m. In this process the gain in surface energy will be :

(A) 2.8×10^{-4} J (B) 1.5×10^{-3} J (C) 1.9×10^{-4} J (D) 9.4×10^{-5} J

Q110. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is :

(A) 19.6 m (B) 29.4 m (C) 39.2 m (D) 73.5 m

Q111. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

The velocity of a small ball of mass 'm' and density d_1, when dropped in a container filled with glycerin, becomes constant after some time. If the density of glycerin is d_2, then the viscous force acting on the ball, will be :

(A) $mg\left(1-\frac{d_1}{d_2}\right)$ (B) $mg\left(1-\frac{d_2}{d_1}\right)$

(C) $mg\left(\frac{d_1}{d_2}-1\right)$ (D) $mg\left(\frac{d_2}{d_1}-1\right)$

Q112. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

If ρ is the density and η is coefficient of viscosity of fluid which flows with a speed v in the pipe of diameter d, the correct formula for Reynolds number R_e is :

(A) $R_e = \frac{\eta d}{\rho v}$ (B) $R_e = \frac{\rho v}{\eta d}$

(C) $R_e = \frac{\rho v d}{\eta}$ (D) $R_e = \frac{\eta}{\rho v d}$

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Q113. JEE Main 2022 (Online) 25th June Morning Shift

The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as :

(A) $r^{1/2}$ (B) r (C) r^2 (D) r^3

Open solution

Q114. JEE Main 2022 (Online) 24th June Morning Shift

The bulk modulus of a liquid is $3 \times 10^{10}\ Nm^{-2}$. The pressure required to reduce the volume of liquid by 2% is :

(A) $3 \times 10^8\ Nm^{-2}$ (B) $9 \times 10^8\ Nm^{-2}$ (C) $6 \times 10^8\ Nm^{-2}$ (D) $12 \times 10^8\ Nm^{-2}$

Open solution

Q115. JEE Main 2021 (Online) 31st August Evening Shift

Four identical hollow cylindrical columns of mild steel support a big structure of mass 50×10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform local distribution, calculate the compression strain of each column. [Use Y = 2.0×10^{11} Pa, g = 9.8 m/s^2]

(A) 3.60×10^{-8} (B) 2.60×10^{-7} (C) 1.87×10^{-3} (D) 7.07×10^{-4}

Open solution

Q116. JEE Main 2021 (Online) 31st August Morning Shift

A uniform heavy rod of weight 10 kg ms^{-2}, cross-sectional area 100 cm^2 and length 20 cm is hanging from a fixed support. Young modulus of the material of the rod is $2 \times 10^{11}\ Nm^{-2}$. Neglecting the lateral contraction, find the elongation of rod due to its own weight.

(A) 2×10^{-9} m (B) 5×10^{-8} m (C) 4×10^{-8} m (D) 5×10^{-10} m

Open solution

Q117. JEE Main 2021 (Online) 27th August Morning Shift

In Millikan's oil drop experiment, what is viscous force acting on an uncharged drop of radius 2.0×10^{-5} m and density $1.2 \times 10^3\ kgm^{-3}$? Take viscosity of liquid = $1.8 \times 10^{-5}\ Nsm^{-2}$. (Neglect buoyancy due to air).

(A) 3.8×10^{-11} N (B) 3.9×10^{-10} N (C) 1.8×10^{-10} N (D) 5.8×10^{-10} N

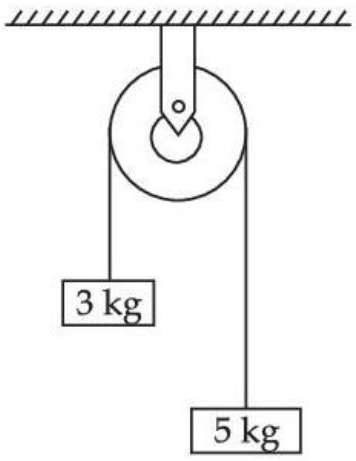
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Q118. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

Two blocks of masses 3 kg and 5 kg are connected by a metal wire going over a smooth pulley. The breaking stress of the metal is $\frac{24}{\pi} \times 10^2$ Nm^{-2}. What is the minimum radius of the wire ? (Take g = 10 ms^{-2})



(A) 125 cm (B) 1250 cm (C) 12.5 cm (D) 1.25 cm

Q119. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube. [Take surface tension of water T = 7.3 $\times$ 10^{-2} Nm^{-1}, angle of contact = 0, g = 10 ms^2 and density of water = 1.0 $\times$ 10^3 kg m^{-3}]

(A) 3.62 mm (B) 2.19 mm (C) 5.34 mm (D) 4.97 mm

Q120. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

A raindrop with radius R = 0.2 mm falls from a cloud at a height h = 2000 m above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is :

[Density of water f_w = 1000 kg m^{-3} and Density of air f_a = 1.2 kg m^{-3}, g = 10 m/s^2, Coefficient of viscosity of air = 1.8 $\times$ 10^{-5} Nsm^{-2}]

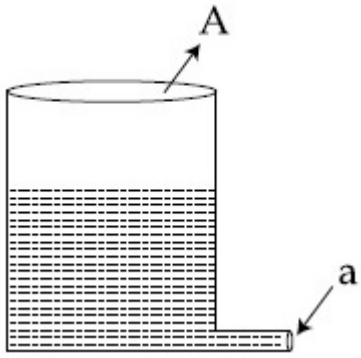
(A) 250.6 ms^{-1} (B) 43.56 ms^{-1} (C) 4.94 ms^{-1} (D) 14.4 ms^{-1}

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Q121. JEE Main 2021 (Online) 27th July Morning Shift

A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is (a < < A) :

Open solution



(A) $\frac{A}{2a}$ (B) None of these (C) $\frac{2a}{A}$ (D) $\frac{a}{A}$

Q122. JEE Main 2021 (Online) 25th July Evening Shift

Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to :

Open solution

(A) $\frac{r_1 r_2}{r_1+r_2}$ (B) $\sqrt{r_1 r_2}$

(C) $\sqrt{r_1^2+r_2^2}$ (D) $\frac{r_1+r_2}{2}$

Q123. JEE Main 2021 (Online) 25th July Morning Shift

Two wires of same length and radius are joined end to end and loaded. The Young's moduli of the materials of the two wires are Y_1 and Y_2. The combination behaves as a single wire then its Young's modulus is :

Open solution

(A) $Y = \frac{2Y_1Y_2}{3(Y_1+Y_2)}$ (B) $Y = \frac{2Y_1Y_2}{Y_1+Y_2}$

(C) $Y = \frac{Y_1Y_2}{2(Y_1+Y_2)}$ (D) $Y = \frac{Y_1Y_2}{Y_1+Y_2}$

Q124. JEE Main 2021 (Online) 20th July Evening Shift

The length of a metal wire is l_1, when the tension in it is T_1 and is l_2 when the tension is T_2. The natural length of the wire is :

Open solution

(A) $\sqrt{l_1 l_2}$ (B) $\frac{l_1T_2-l_2T_1}{T_2-T_1}$

(C) $\frac{l_1T_2+l_2T_1}{T_2+T_1}$ (D) $\frac{l_1+l_2}{2}$

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Q125. JEE Main 2021 (Online) 20th July Evening Shift

Two small drops of mercury each of radius R coalesce to form a single large drop. The ratio of total surface energy before and after the change is :

(A) $2^{\frac{1}{3}} : 1$ (B) $1 : 2^{\frac{1}{3}}$ (C) 2 : 1 (D) 1 : 2

Open solution

Q126. JEE Main 2021 (Online) 20th July Morning Shift

The value of tension in a long thin metal wire has been changed from T_1 to T_2. The lengths of the metal wire at two different values of tension T_1 and T_2 are l_1 and l_2 respectively. The actual length of the metal wire is :

(A) $\frac{l_1+l_2}{2}$ (B) $\sqrt{T_1T_2l_1l_2}$

(C) $\frac{T_1l_2-T_2l_1}{T_1-T_2}$ (D) $\frac{T_1l_1-T_2l_2}{T_1-T_2}$

Open solution

Q127. JEE Main 2021 (Online) 17th March Evening Shift

An object is located at 2 km beneath the surface of the water. If the fractional compression $\frac{\Delta V}{V}$ is 1.36%, the ratio of hydraulic stress to the corresponding hydraulic strain will be _____. [Given : density of water is 1000 kgm^{-3} and g = 9.8 ms^{-2}]

(A) 1.44×10^7 Nm^{-2} (B) 1.44×10^9 Nm^{-2}

(C) 1.96×10^7 Nm^{-2} (D) 2.26×10^9 Nm^{-2}

Open solution

Q128. JEE Main 2021 (Online) 17th March Morning Shift

When two soap bubbles of radii a and b (b > a) coalesce, the radius of curvature of common surface is :

(A) $\frac{b-a}{ab}$ (B) $\frac{a+b}{ab}$ (C) $\frac{ab}{a+b}$ (D) $\frac{ab}{b-a}$

Open solution

Q129. JEE Main 2021 (Online) 16th March Evening Shift

What will be the nature of flow of water from a circular tap, when its flow rate increased from 0.18 L/min to 0.48 L/min? The radius of the tap and viscosity of water are 0.5 cm and 10^{-3} Pa s, respectively. (Density of water : 10^3 kg/m^3)

(A) Steady flow to unsteady flow (B) Unsteady to steady flow

(C) Remains turbulent flow (D) Remains steady flow

Open solution

Q130. JEE Main 2021 (Online) 16th March Morning Shift

The pressure acting on a submarine is 3×10^5 Pa at a certain depth. If the depth is doubled, the percentage increase in the pressure acting on the submarine would be :

(Assume that atmospheric pressure is 1×10^5 Pa and density of water is 10^3 kg m^{-3}, g = 10 ms^{-2})

(A) $\frac{200}{5}$ % (B) $\frac{200}{3}$ % (C) $\frac{3}{200}$ % (D) $\frac{5}{200}$ %

Open solution

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Q131. JEE Main 2021 (Online) 26th February Evening Shift

The length of metallic wire is l_1 when tension in it is T_1. It is l_2 when the tension is T_2. The original length of the wire will be :

(A) $\frac{T_1 l_1 - T_2 l_2}{T_2 - T_1}$

(B) $\frac{l_1 + l_2}{2}$

(C) $\frac{T_2 l_1 + T_1 l_2}{T_1 + T_2}$

(D) $\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$

Open solution

Q132. JEE Main 2021 (Online) 26th February Morning Shift

A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be :

(A) $\frac{2T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$

(B) $\frac{3T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$

(C) $\frac{3T}{rJ}$

(D) $\frac{2T}{rJ}$

Open solution

Q133. JEE Main 2021 (Online) 26th February Morning Shift

The normal density of a material is ρ and its bulk modulus of elasticity is K. The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be :

(A) $\frac{\rho K}{P}$

(B) $\frac{PK}{\rho}$

(C) $\frac{\rho P}{K}$

(D) $\frac{K}{\rho P}$

Open solution

Q134. JEE Main 2021 (Online) 24th February Morning Shift

If Y, K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

(A) $Y = \frac{9K\eta}{3K - \eta} N/m^2$

(B) $Y = \frac{9K\eta}{2\eta + 3K} N/m^2$

(C) $\eta = \frac{3YK}{9K + Y} N/m^2$

(D) $K = \frac{Y\eta}{9\eta - 3Y} N/m^2$

Open solution

Q135. JEE Main 2020 (Online) 6th September Evening Slot

A fluid is flowing through a horizontal pipe of varying cross-section, with speed v ms^{-1} at a point where the pressure is P pascal. At another point where pressure is $\frac{P}{2}$ Pascal its speed is V ms^{-1}. If the density of the fluid is ρ kg m^{-3} and the flow is streamline, then V is equal to :

(A) $\sqrt{\frac{P}{2\rho} + v^2}$

(B) $\sqrt{\frac{P}{\rho} + v^2}$

(C) $\sqrt{\frac{2P}{\rho} + v^2}$

(D) $\sqrt{\frac{P}{\rho} + v}$

Open solution

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Q136. JEE Main 2020 (Online) 6th September Morning Slot

An object of mass m is suspended at the end of a massless wire of length L and area of crosssection A. Young modulus of the material of the wire is Y. If the mass is pulled down slightly its frequency of oscillation along the vertical direction is :

Open solution

(A) $f = \frac{1}{2\pi}\sqrt{\frac{YA}{mL}}$ (B) $f = \frac{1}{2\pi}\sqrt{\frac{mL}{YA}}$

(C) $f = \frac{1}{2\pi}\sqrt{\frac{YL}{mA}}$ (D) $f = \frac{1}{2\pi}\sqrt{\frac{mA}{YL}}$

Q137. JEE Main 2020 (Online) 5th September Evening Slot

In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to :
(ignore viscosity of air)

Open solution

(A) r (B) r^4 (C) r^3 (D) r^2

Q138. JEE Main 2020 (Online) 5th September Morning Slot

A hollow spherical shell of outer radius R floats just submerged under the water surface. The inner radius of the shell is r. If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is :

Open solution

(A) $\frac{2}{3}$R (B) $\frac{4}{9}$R (C) $\frac{1}{3}$R (D) $\frac{8}{9}$R

Q139. JEE Main 2020 (Online) 4th September Evening Slot

A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to :
(Given bulk modulus of metal, B = 8×10^{10} Pa)

Open solution

(A) 0.6 (B) 20 (C) 1.67 (D) 5

Q140. JEE Main 2020 (Online) 4th September Evening Slot

Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2 . When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is:

Open solution

(A) $gdS(x_2 + x_1)^2$ (B) $gdS\left(x_2^2 + x_1^2\right)$

(C) $\frac{1}{4}gdS(x_2 - x_1)^2$ (D) $\frac{3}{4}gdS(x_2 - x_1)^2$

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Q141. JEE Main 2020 (Online) 4th September Morning Slot

A air bubble of radius 1 cm in water has an upward acceleration 9.8 cm s^{-2}. The density of water is 1 gm cm^{-3} and water offers negligible drag force on the bubble. The mass of the bubble is (g = 980 cm/s^2).

Open solution

(A) 1.52 gm (B) 4.51 gm (C) 3.15 gm (D) 4.15 gm

Q142. JEE Main 2020 (Online) 3rd September Morning Slot

Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is :

Open solution

(A) 4 : 1 (B) 8 : 1 (C) 2 : 1 (D) 0.8 : 1

Q143. JEE Main 2020 (Online) 2nd September Evening Slot

A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm^{-1}, density = 667 kg m^{-3}) which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then h is close to (g = 10 ms^{-2})

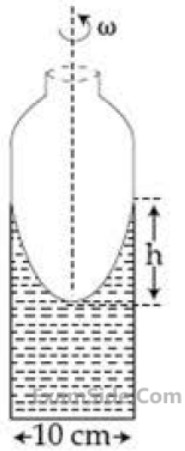
Open solution

(A) 0.049 m (B) 0.087 m (C) 0.137 m (D) 0.172 m

Q144. JEE Main 2020 (Online) 2nd September Morning Slot

A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 cm and the angular speed of rotation is ω rad s^{-1}. The difference in the height, h (in cm) of liquid at the centre of vessel and at the side will be :

Open solution



(A) $\frac{2\omega^2}{25g}$ (B) $\frac{5\omega^2}{2g}$ (C) $\frac{25\omega^2}{2g}$ (D) $\frac{2\omega^2}{5g}$

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Q145. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet) :

(A) $r = \sqrt{\frac{T}{(d-\rho)g}}$

(B) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$

(C) $r = \sqrt{\frac{T}{(d+\rho)g}}$

(D) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$

Q146. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

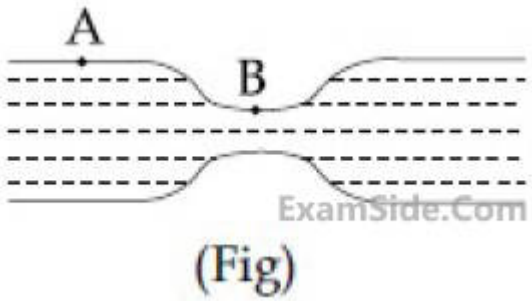
Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is:

(A) 1 : 2 (B) 2 : 1 (C) $1 : \sqrt{2}$ (D) $\sqrt{2} : 1$

Q147. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

Water flows in a horizontal tube (see figure).
The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2, respectively. Find the rate of flow of water through the tube.
(density of water = 1000 kgm^{-3})



(Fig)

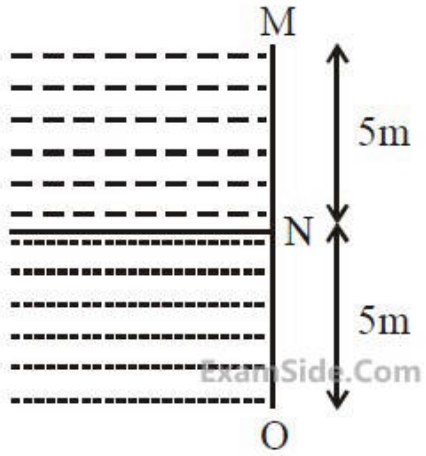
(A) 1810 cm^3/s (B) 2420 cm^3/s (C) 3020 cm^3/s (D) 2720 cm^3/s

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Q148. JEE Main 2020 (Online) 8th January Evening Slot

Two liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing)

Open solution



(A) 1/3 (B) 1/2 (C) 1/4 (D) 2/3

Q149. JEE Main 2020 (Online) 8th January Morning Slot

Consider a solid sphere of radius R and mass density
$\rho(r) = \rho_0\left(1 - \frac{r^2}{R^2}\right), 0 < r \leq R$
The minimum density of a liquid in which it will float is :

Open solution

(A) $\frac{2\rho_0}{3}$ (B) $\frac{2\rho_0}{5}$ (C) $\frac{\rho_0}{5}$ (D) $\frac{\rho_0}{3}$

Q150. JEE Main 2020 (Online) 8th January Morning Slot

A leak proof cylinder of length 1m, made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm. When the temperature of water is increased to 4°C, the height of the cylinder above the water surface becomes 21 cm. The density of water at T = 4°C, relative to the density at T = 0°C is close to :

Open solution

(A) 1.04 (B) 1.26 (C) 1.01 (D) 1.03

Q151. JEE Main 2020 (Online) 7th January Evening Slot

An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is :

Open solution

(A) $\frac{3}{4}$ (B) $\frac{9}{16}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{81}{256}$

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Q152. JEE Main 2019 (Online) 12th April Evening Slot

A solid sphere, of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity . The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, v_2, when falling through the same fluid, the ratio (v_1/v_2) equals :

Open solution

(A) $\frac{1}{9}$ (B) $\frac{1}{27}$ (C) 27 (D) 9

Q153. JEE Main 2019 (Online) 12th April Evening Slot

The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-\alpha r^4}$. Then the total number of molecules is proportional to :

Open solution

(A) $n_0\alpha^{-3/4}$ (B) $n_0\alpha^{-3}$ (C) $n_0\alpha^{1/4}$ (D) $\sqrt{n_0}\alpha^{1/2}$

Q154. JEE Main 2019 (Online) 12th April Evening Slot

A uniform cylindrical rod of length L and radius r, is made from a material whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equal to :

Open solution

(A) $\frac{3F}{(\pi r^2 YT)}$ (B) $\frac{6F}{(\pi r^2 YT)}$

(C) $\frac{F}{(3\pi r^2 YT)}$ (D) $9F\left(\pi r^2 YT\right)$

Q155. JEE Main 2019 (Online) 10th April Evening Slot

Water from a tap emerges vertically downwards with an initial speed of 1.0 ms^{-1} . The cross-sectional area of the tap is 10^{-4} m^2. Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 m below the tap would be : (Take g = 10 ms^{-2})

Open solution

(A) 5×10^{-4} m^2 (B) 2×10^{-5} m^2 (C) 5×10^{-5} m^2 (D) 1×10^{-5} m^2

Q156. JEE Main 2019 (Online) 10th April Evening Slot

A submarine experiences a pressure of 5.05×10^6 Pa at a depth of d_1 in a sea. When it goes further to a depth of d_2, it experiences a pressure of 8.08×10^6 Pa. Then $d_2 - d_1$ is approximately (density of water = 10^3 kg/m^3 and acceleration due to gravity = 10 ms^{-2}) :

Open solution

(A) 600 m (B) 400 m (C) 300 m (D) 500 m

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Q157. JEE Main 2019 (Online) 10th April Evening Slot

In an experiment, brass and steel wires of length 1 m each with areas of cross section 1mm^2 are used. The wires are connected in series and one end of the combined wire is connected to a rigid support and other end is subjected to elongation. The stress required to produce a net elongation of 0.2 mm is, [Given, the Young's Modulus for steel and brass are, respectively, 120×10^9 N/m^2 and 60×10^9 N/m^2]

Open solution

(A) 8.0×10^6 N/m^2 (B) 1.2×10^6 N/m^2 (C) 0.2×10^6 N/m^2 (D) 1.8×10^6 N/m^2

Q158. JEE Main 2019 (Online) 10th April Evening Slot

A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water? [Take, density of water = 10^3 kg/m^3]

Open solution

(A) 30.1 kg (B) 87.5 kg (C) 65.4 kg (D) 46.3 kg

Q159. JEE Main 2019 (Online) 10th April Evening Slot

The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass rod if it is to support a 400 N load without exceeding its elastic limit?

Open solution

(A) 1.16 mm (B) 1.36 mm (C) 1.00 mm (D) 0.90 mm

Q160. JEE Main 2019 (Online) 10th April Morning Slot

The ratio of surface tensions of mercury and water is given to be 7.5 while the ratio of their densities is 13.6. Their contact angles, with glass, are close to 135° and 0°, respectively. It is observed that mercury gets depressed by an amount h in a capillary tube of radius r_1, while water rises by the same amount h in a capillary tube of radius r_2. The ratio, (r_1/r_2), is then close to :

Open solution

(A) 2/5 (B) 2/3 (C) 3/5 (D) 4/5

Q161. JEE Main 2019 (Online) 9th April Evening Slot

A wooden block floating in a bucket of water has 4/5 of its volume submerged. When certain amount of an oil is poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is :-

Open solution

(A) 0.8 (B) 0.7 (C) 0.6 (D) 0.5

Q162. JEE Main 2019 (Online) 9th April Morning Slot

If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is :

Open solution

(A) M (B) 4M (C) M/2 (D) 2M

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Q163. JEE Main 2019 (Online) 8th April Evening Slot

Young's moduli of two wires A and B are in the ratio 7 : 4. Wire A is 2 m long and has radius R. Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R is close to :-

Open solution

(A) 1.7 mm (B) 1.9 mm (C) 1.3 mm (D) 1.5 mm

Q164. JEE Main 2019 (Online) 8th April Morning Slot

Water from a pipe is coming at a rate of 100 litres per minute. If the radius of the pipe is 5 cm, the Reynolds number for the flow is of the order of : (density of water = 1000 kg/m^3, coefficient of viscosity of water = 1mPas)

Open solution

(A) 10^6 (B) 10^4 (C) 10^3 (D) 10^2

Q165. JEE Main 2019 (Online) 8th April Morning Slot

A boy's catapult is made of rubber cord which is 42 cm long, with 6 mm diameter of cross-section and of negligible mass. The boy keeps a stone weighing 0.02kg on it and stretches the cord by 20 cm by applying a constant force. When released, the stone flies off with a velocity of 20 ms^{-1}. Neglect the change in the area of cross-section of the cord while stretched. The Young's modulus of rubber is closest to:

Open solution

(A) 10^4 Nm^{-2} (B) 10^6 Nm^{-2} (C) 10^8 Nm^{-2} (D) 10^3 Nm^{-2}

Q166. JEE Main 2019 (Online) 8th April Morning Slot

A steel wire having a radius of 2.0 mm, carrying a load of 4 kg, is hanging from a ceiling. Given that g = 3.1π ms^{-2}, what will be the tensile stress that would be developed in the wire ?

Open solution

(A) 3.1×10^6 Nm^{-2} (B) 6.2×10^6 Nm^{-2} (C) 4.8×10^6 Nm^{-2} (D) 5.2×10^6 Nm^{-2}

Q167. JEE Main 2019 (Online) 12th January Evening Slot

A load of mass M kg is suspended from a steel wire of length 2m and radius 1.0 mm in Searle's apparatus experiment. The increase in length produced in the wire is 4.0 mm. Now the load is fully immersed in a liquid of relative density 2. The relative density of the material of load is 8.

Open solution

The new value of increase in length of the steel wire is:

(A) 5.0 mm (B) zero (C) 3.0 mm (D) 4.0 mm

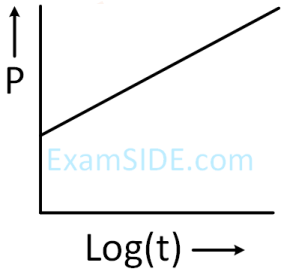
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Q168. JEE Main 2019 (Online) 12th January Evening Slot

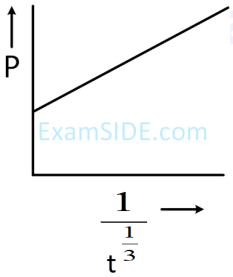
Open solution

A soap bubble, blown by a mechanical pump at the mouth of a tube, increases in volume, with time, at a constant rate. The graph that correctly depicts the time dependence of pressure inside the bubble is given by :

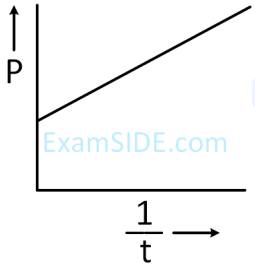
(A)



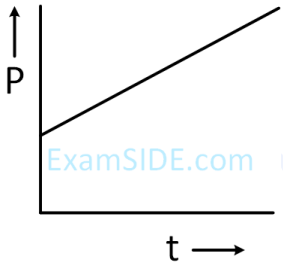
(B)



(C)



(D)



Q169. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be :

(A) 2.0 (B) 1.2 (C) 0.1 (D) 0.4

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Q170. JEE Main 2019 (Online) 11th January Morning Slot

A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% looses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be :

Open solution

(A) $\frac{3}{4}\rho v^2$ (B) $\frac{1}{4}\rho v^2$ (C) $\frac{1}{2}\rho v^2$ (D) ρv^2

Q171. JEE Main 2019 (Online) 10th January Morning Slot

Water flows into a large tank with flat bottom at the rate of $10^{-4}m^3s^{-1}$. Water is also leaking out of a hole ofarea 1 cm^2 at its bottom. If the height of the water in the tank remains steady, then this height is -

Open solution

(A) 2.9 cm (B) 5.1 cm (C) 4 cm (D) 1.7 cm

Q172. JEE Main 2019 (Online) 9th January Evening Slot

The top of a water tank is open to air and its water level is mainted. It is giving out 0.74 m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the center of the opening from the level of water in the tank is close to :

Open solution

(A) 6.0 m (B) 4.8 m (C) 9.6 m (D) 2.9 m

Q173. JEE Main 2018 (Online) 16th April Morning Slot

A small soap bubble of radius 4 cm is trapped inside another bubble of radius 6 cm without any contact. Let P_2 be the pressure inside the inner bubble and P_0, the pressure outside the outer bubble. Radius of another bubble with pressure difference $P_2 - P_0$ between its inside and outside would be :

Open solution

(A) 12 cm (B) 2.4 cm (C) 6 cm (D) 4.8 cm

Q174. JEE Main 2018 (Offline)

A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross section of cylindrical container. When a mass m is placed on the surface of the piston to compress the liquid, the fractional decrement in the radius of the sphere, $\left(\frac{dr}{r}\right)$ is:

Open solution

(A) $\frac{mg}{Ka}$ (B) $\frac{Ka}{mg}$ (C) $\frac{Ka}{3mg}$ (D) $\frac{mg}{3Ka}$

Q175. JEE Main 2018 (Online) 15th April Evening Slot

When an air bubble of radius r rises from the bottom to the surface of a lake its radius becomes $\frac{5r}{4}$. Taking the atmospheric pressure to be equal to 10 m height of water column, the depth of the lake would approximately be (ignore the surface tension and the effect of temperature) :

Open solution

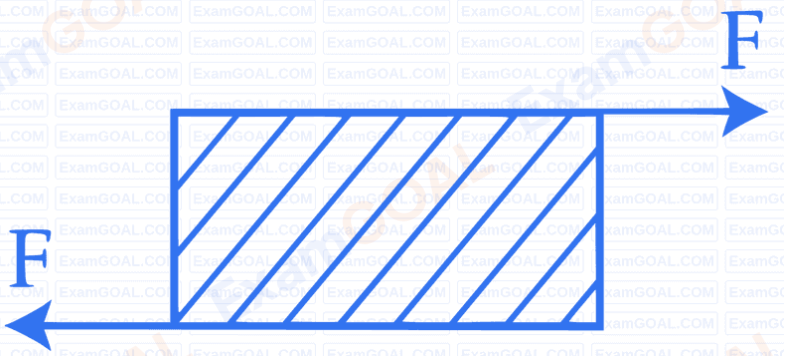
(A) 11.2 m (B) 8.7 m (C) 9.5 m (D) 10.5 m

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Q176. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

As shown in the figure, forces of 10^5 N each are applied in opposite directions, on the upper and lower faces of a cube of side 10 cm, shifting the upper face parallel to itself by 0.5 cm. If the side of another cube of the same material is 20 cm, then under similar conditions as above, the displacement will be :



(A) 0.25 cm (B) 0.37 cm (C) 0.75 cm (D) 1.00 cm

Q177. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

A thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are ρ_1 and ρ_2 $(\rho_1 > \rho_2)$, fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is :

(A) $\theta = \tan^{-1}\pi\left(\frac{\rho_1}{\rho_2}\right)$

(B) $\theta = \tan^{-1}\frac{\pi}{2}\left(\frac{\rho_1}{\rho_2}\right)$

(C) $\theta = \tan^{-1}\left(\frac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)$

(D) $\theta = \tan^{-1}\frac{\pi}{2}\left(\frac{\rho_1+\rho_2}{\rho_1-\rho_2}\right)$

Q178. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

Two tubes of radii r_1 and r_2, and lengths l_1 and l_2 , respectively, are connected in series and a liquid flows through each of them in stream line conditions. P_1 and P_2 are pressure differences across the two tubes.

If P_2 is $4P_1$ and l_2 is $\frac{l_1}{4}$, then the radius r_2 will be equal to :

(A) r_1 (B) $2r_1$ (C) $4r_1$ (D) $\frac{r_1}{2}$

Q179. JEE Main 2017 (Offline)

Open solution

A man grows into a giant such that his linear dimensions increase by a factor of 9. Assuming that his density remains same, the stress in the leg will change by a factor of :

(A) $\frac{1}{81}$ (B) 9 (C) $\frac{1}{9}$ (D) 81

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Q180. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

A thin 1 m long rod has a radius of 5 mm. A force of 50 πkN is applied at one end to determine its Young's modulus. Assume that the force is exactly known. If the least count in the measurement of all lengths is 0.01 mm, which of the following statements is **false** ?

(A) $\frac{\Delta Y}{Y}$ gets minimum contribution from the uncertainty in the length.

(B) The figure of merit is the largest for the length of the rod.

(C) The maximum value of Y that can be determined is 10^{14} N/m^2

(D) $\frac{\Delta Y}{Y}$ gets its maximum contribution from the uncertainty in strain

Q181. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

A bottle has an opening of radius a and length b. A cork of length b and radius (a + Δa) where (Δa < < a) is compressed to fit into the opening completely (See figure). If the bulk modulus of cork is B and frictional coefficient between the bottle and cork is μ then the force needed to push the cork into the bottle is :



(A) (π μ B b) Δa

(B) (2π μ B b) Δa

(C) (π μ B b) a

(D) (4π μ B b) Δa

Q182. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

A uniformly tapering conical wire is made from a material of Young's modulus Y and has a normal, unextended length L. The radii, at the upper and lower ends of this conical wire, have values R and 3 R, respectively. The upper end of the wire is fixed to a rigid support and a mass M is suspended from its lower end. The equilibrium extended length, of this wire, would equal :

(A) L $\left(1+\frac{2}{9}\frac{Mg}{\pi YR^2}\right)$

(B) L $\left(1+\frac{1}{3}\frac{Mg}{\pi YR^2}\right)$

(C) L $\left(1+\frac{1}{9}\frac{Mg}{\pi YR^2}\right)$

(D) L $\left(1+\frac{2}{3}\frac{Mg}{\pi YR^2}\right)$

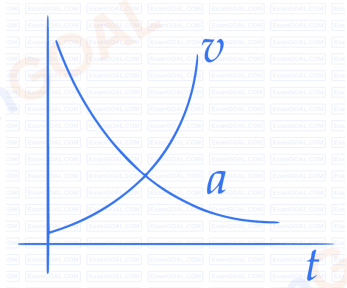
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Q183. JEE Main 2016 (Online) 9th April Morning Slot

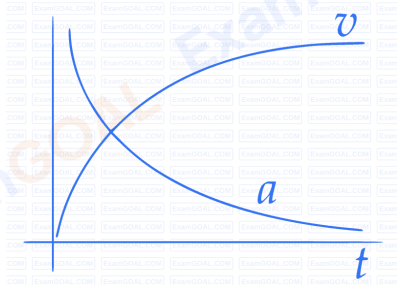
Open solution

Which of the following option correctly describes the variation of the speed *v* and acceleration *'a'* of a point mass falling vertically in a viscous medium that applies a force F = − *kv,* where 'k' is a constant, on the body ? (Graphs are schematic and not drawn to scale)

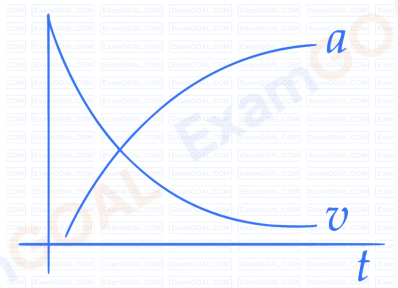
(A)



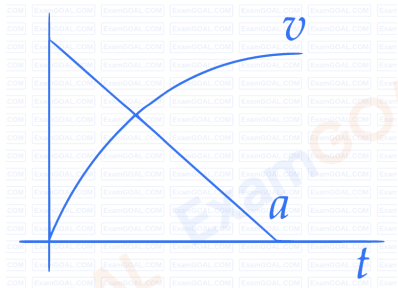
(B)



(C)



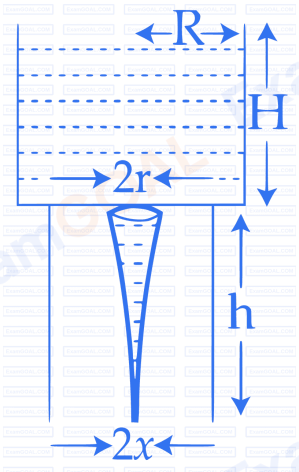
(D)



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Q184. JEE Main 2016 (Online) 9th April Morning Slot

Open solution



Consider a water jar of radius R that has water filled up to height H and is kept on astand of height h (see figure). Through a hole of radius r (r << R) at its bottom, the water leaks out and the stream of water coming down towards the ground has a shape like a funnel as shown in the figure. If the radius of the cross-section of water stream when it hits the ground is x. Then :

(A) $x = r\left(\frac{H}{H+h}\right)$

(B) $x = r\left(\frac{H}{H+h}\right)^{\frac{1}{2}}$

(C) $x = r\left(\frac{H}{H+h}\right)^{\frac{1}{4}}$

(D) $x = r\left(\frac{H}{H+h}\right)^{2}$

Q185. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

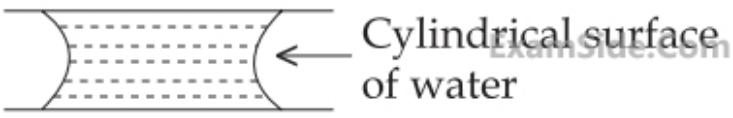
If it takes 5 minutes to fill a 15 litre bucket from a water tap of diameter $\frac{2}{\sqrt{\pi}}$ cm then the Reynolds number for the flow is (density of water = 10^3 kg/m^3 and viscosity of water = 10^{-3} Pa.s) close to :

(A) 5500 (B) 11,000 (C) 550 (D) 1100

Q186. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

If two glass plates have water between them and are separated by very small distance (see figure), it is very difficult to pull them apart. It is because the water in between forms cylindrical surface on the side that gives rise to lower pressure in the water in comparison to atmosphere. If the radius of the cylindrical surface is R and surface tension of water is T then the pressure in water between the plates is lower by :



(A) $\frac{T}{4R}$ (B) $\frac{T}{2R}$ (C) $\frac{4T}{R}$ (D) $\frac{2T}{R}$

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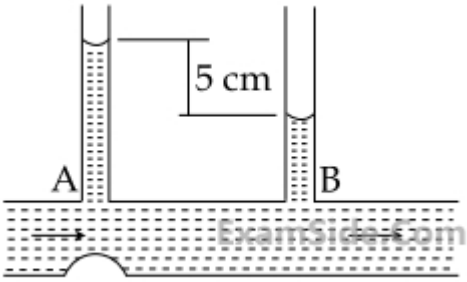
Q187. JEE Main 2014 (Online) 19th April Morning Slot

The velocity of water in a river is 18 km/hr near the surface. If the river is 5 m deep, find the shearing stress between the horizontal layers of water. The co-efficient of viscosity of water = 10^{-2} poise.

(A) 10^{-1} N/m^2 (B) 10^{-2} N/m^2 (C) 10^{-3} N/m^2 (D) 10^{-4} N/m^2

Open solution

Q188. JEE Main 2014 (Online) 19th April Morning Slot



Open solution

In the diagram shown, if the difference in the two tubes of the manometer is 5 cm, the cross section of the tube at A and B is 6 mm^2 and 10 mm^2 respectively. The rate at which water flows through the tube (g = 10 ms^{-2})

(A) 7.5 cc/s (B) 8.0 cc/s (C) 10.0 cc/s (D) 12.5 cc/s

Q189. JEE Main 2014 (Online) 19th April Morning Slot

A large number of liquid drops each of radius r coalesce to form a single drop of radius R. The energy released in the process is converted into kinetic energy of the big drop so formed. The speed of the big drop is (given surface tension of liquid T, density ρ)

(A) $\sqrt{\frac{T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$ (B) $\sqrt{\frac{2T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

(C) $\sqrt{\frac{4T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$ (D) $\sqrt{\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

Open solution

Q190. JEE Main 2014 (Online) 12th April Morning Slot

Steel ruptures when a shear of 3.5×10^8 N m^{-2} is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly :

(A) 1.4×10^4 N (B) 2.7×10^4 N (C) 3.3×10^4 N (D) 1.1×10^4 N

Open solution

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Q191. JEE Main 2014 (Online) 12th April Morning Slot

A cylindrical vessel of cross-section A contains water to a height h. There is a hole in the bottom of radius 'a'. The time in which it will be emptied is :

(A) $\frac{2A}{\pi a^2}\sqrt{\frac{h}{g}}$ (B) $\frac{\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$

(C) $\frac{2\sqrt{2}A}{\pi a^2}\sqrt{\frac{h}{g}}$ (D) $\frac{A}{\sqrt{2}\,\pi a^2}\sqrt{\frac{h}{g}}$

Open solution

Q192. JEE Main 2014 (Online) 12th April Morning Slot

Two soap bubbles coalesce to form a single bubble. If V is the subsequent change in volume of contained air and S the change in total surface area, T is the surface tension and P atmospheric pressure, which of the following relation is **correct** ?

(A) 4PV + 3ST = 0 (B) 3PV + 4ST = 0 (C) 2PV + 3ST = 0 (D) 3PV + 2ST = 0

Open solution

Q193. JEE Main 2014 (Online) 11th April Morning Slot

The Bulk moduli of Ethanol, Mercury and water are given as 0.9, 25 and 2.2 respectively in units of 10^9 Nm^{-2}. For a given value of pressure, the fractional compression in volume is $\frac{\Delta V}{V}$. Which of the following statements about $\frac{\Delta V}{V}$ for these three liquids is correct ?

(A) Ethanol > Water > Mercury (B) Water > Ethanol > Mercury

(C) Mercury > Ethanol > Water (D) Ethanol > Mercury > Water

Open solution

Q194. JEE Main 2014 (Online) 11th April Morning Slot

A tank with a small hole at the bottom has been filled with water and kerosene (specific gravity 0.8). The height of water is 3 m and that of kerosene 2 m. When the hole is opened the velocity of fluid coming out from it is nearly : (take g = 10 ms^{-2} and density of water = 10^3 kg m^{-3})

(A) 10.7 ms^{-1} (B) 9.6 ms^{-1} (C) 8.5 ms^{-1} (D) 7.6 ms^{-1}

Open solution

Q195. JEE Main 2014 (Online) 11th April Morning Slot

An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m^3. The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth is the bubble below the surface of the liquid ? (g = 9.8 ms^{-2})

(A) 0.1 m (B) 0.15 m (C) 0.20 m (D) 0.25 m

Open solution

Q196. JEE Main 2014 (Online) 9th April Morning Slot

In materials like aluminium and copper, the correct order of magnitude of various elastic moduli is :

(A) Young's moduli < shear moduli < bulk moduli.

(B) Bulk moduli < shear moduli < Young's moduli.

(C) Shear moduli < Young's moduli < bulk moduli.

(D) Bulk moduli < Young's moduli < shear moduli

Open solution

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Q197. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A capillary tube is immersed vertically in water and the height of the water column is x. When this arrangement is taken into a mine of depth d, the height of the water column is y. If R is the radius of earth, the ratio $\frac{x}{y}$ is :

(A) $\left(1-\frac{d}{R}\right)$ (B) $\left(1-\frac{2d}{R}\right)$

(C) $\left(\frac{R-d}{R+d}\right)$ (D) $\left(\frac{R+d}{R-d}\right)$

Q198. JEE Main 2014 (Offline)

Open solution

An open glass tube is immersed in mercury in such a way that a length of $8\ cm$ extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional $46\ cm$. What will be length of the air column above mercury in the tube now? (Atmospheric pressure $= 76\ cm$ of Hg)

(A) $16\ cm$ (B) $22\ cm$ (C) $38\ cm$ (D) $6\ cm$

Q199. JEE Main 2014 (Offline)

Open solution

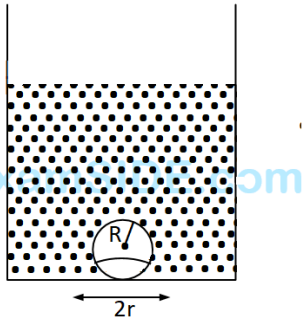
The pressure that has to be applied to the ends of a steel wire of length $10\ cm$ to keep its length constant when its temperature is raised by $100^{\circ}C$ is: (For steel Young's modulus is $2\times10^{11}\ Nm^{-2}$ and coefficient of thermal expansion is $1.1\times10^{-5}\ K^{-1}$)

(A) $2.2\times10^{8}\ Pa$ (B) $2.2\times10^{9}\ Pa$ (C) $2.2\times10^{7}\ Pa$ (D) $2.2\times10^{6}\ Pa$

Q200. JEE Main 2014 (Offline)

Open solution

On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r << R$ and the surface tension of water is T, value of r just before bubbles detach is: (density of water is ρ_w)



(A) $R^2\sqrt{\frac{\rho_w g}{3T}}$ (B) $R^2\sqrt{\frac{\rho_w g}{6T}}$

(C) $R^2\sqrt{\frac{\rho_w g}{T}}$ (D) $R^2\sqrt{\frac{2\rho_w g}{3T}}$

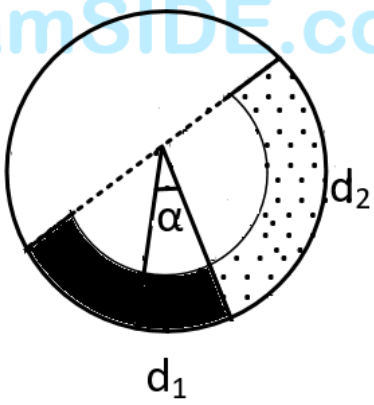
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Q201. JEE Main 2014 (Offline)

There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at center. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :

Open solution

ExamSIDE.com



(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$
(B) $\frac{1+\cos\alpha}{1-\cos\alpha}$
(C) $\frac{1+\tan\alpha}{1-\tan\alpha}$
(D) $\frac{1+\sin\alpha}{1-\cos\alpha}$

Q202. JEE Main 2013 (Online) 25th April Morning Slot

In an experiment, a small steel ball falls through a liquid at a constant speed of 10 cm/s. If the steel ball is pulled upward with a force equal to twice its effective weight, how fast will it move upward?

Open solution

(A) 5 cm/s (B) Zero (C) 10 cm/s (D) 20 cm/s

Q203. JEE Main 2013 (Online) 23th April Morning Slot

A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's moduli of copper and steel are respectively 1.0×10^{11} Nm^{-2} and 2.0×10^{11} Nm^{-2}, the total extension of the composite wire is :

Open solution

(A) 1.75 mm (B) 2.0 mm (C) 1.50 mm (D) 1.25 mm

Q204. JEE Main 2013 (Online) 23th April Morning Slot

Wax is coated on the inner wall of a capillary tube and the tube is then dipped in water. Then, compared to the unwaxed capillary, the angle of contact θ and the height h upto which water rises change. These changes are :

Open solution

(A) θ increases and h also increases
(B) θ decreases and h also decreases
(C) θ increases and h decreases
(D) θ decreases and h increases

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Q205. JEE Main 2013 (Online) 22th April Morning Slot

A uniform wire (Young's modulus 2×10^{11} Nm^{-2}) is subjected to longitudinal tensile stress of 5×10^{7} Nm^{-2}. If the overall volume change in the wire is 0.02%, the fractional decrease in the radius of the wire is close to:

Open solution

(A) 1.0×10^{-4} (B) 1.5×10^{-4} (C) 0.25×10^{-4} (D) 5×10^{-4}

Q206. JEE Main 2013 (Online) 22th April Morning Slot

Air of density 1.2 kg m^{-3} is blowing across the horizontal wings of an aeroplane in such a way that its speeds above and below the wings are 150 ms^{-1} and 100 ms^{-1}, respectively. The pressure difference between the upper and lower sides of the wings, is :

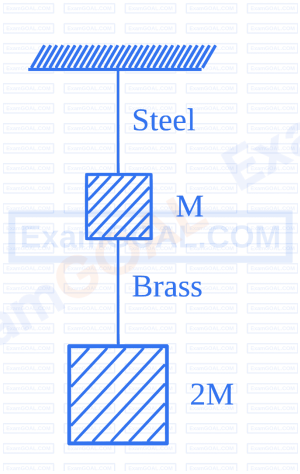
Open solution

(A) 60 Nm^{-2} (B) 180 Nm^{-2} (C) 7500 Nm^{-2} (D) 12500 Nm^{-2}

Q207. JEE Main 2013 (Online) 9th April Morning Slot

If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are a, b and c respectively, then the corresponding ratio of increase in their lengths is :

Open solution



(A) $\frac{3c}{2ab^2}$ (B) $\frac{2a^2c}{b}$ (C) $\frac{3a}{2b^2c}$ (D) $\frac{2ac}{b^2}$

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Q208. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement-1: A capillary is dipped in a liquid and liquid rises to a height h in it. As the temperature of the liquid is raised, the height h increases (if the density of the liquid and the angle of contact remain the same).
Statement-2: Surface tension of a liquid decreases with the rise in its temperature.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation for Statement-1.

(B) Statement-1 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is false.

(D) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation for Statement-1.

Q209. JEE Main 2013 (Offline)

Open solution

A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a mass-less spring such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is:

(A) $\frac{Mg}{k}$

(B) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$

(C) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$

(D) $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

Q210. JEE Main 2013 (Offline)

Open solution

Assume that a drop of liquid evaporates by decreases in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible ? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporization.

(A) $\rho L/T$

(B) $\sqrt{T/\rho L}$

(C) $T/\rho L$

(D) $2T/\rho L$

Q211. AIEEE 2012 (Online) 26th May

Open solution

The terminal velocity of a small sphere of radius a in a viscous liquid is proportional to

(A) a^2

(B) a^3

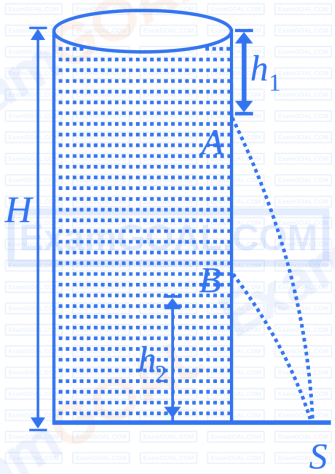
(C) a

(D) a^{-1}

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Q212. AIEEE 2012 (Online) 26th May

In a cylindrical water tank, there are two small holes A and B on the wall at a depth of h_1 from the surface of water and at a height of h_2 from the bottom of water tank. Surface of water is at height H from the bottom of water tank. Water coming out from both holes strikes the ground at the same point S. Find the ratio of h_1 and h_2



Open solution

(A) Depends on H (B) $1:1$ (C) $2:2$ (D) $1:2$

Q213. AIEEE 2012 (Online) 19th May

A steel wire can sustain $100\ \text{kg}$ weight without breaking. If the wire is cut into two equal parts, each part can sustain a weight of

(A) $50\ \text{kg}$ (B) $400\ \text{kg}$ (C) $100\ \text{kg}$ (D) $200\ \text{kg}$

Open solution

Q214. AIEEE 2012 (Online) 19th May

A large number of droplets, each of radius, r coalesce to form a bigger drop of radius, R. An engineer designs a machine so that the energy released in this process is converted into the kinetic energy of the drop. Velocity of the drop is ($T =$ surface tension, $\rho =$ density)

(A) $\left[\frac{T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$ (B) $\left[\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$

(C) $\left[\frac{3T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$ (D) $\left[\frac{2T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)\right]^{1/2}$

Open solution

Q215. AIEEE 2012 (Online) 12th May

Water is flowing through a horizontal tube having cross-sectional areas of its two ends being A and A' such that the ratio A/A' is 5. If the pressure difference of water between the two ends is $3\times10^5\ \text{N m}^{-2}$, the velocity of water with which it enters the tube will be (neglect gravity effects)

(A) $5\ \text{m s}^{-1}$ (B) $10\ \text{m s}^{-1}$ (C) $25\ \text{m s}^{-1}$ (D) $50\sqrt{10}\ \text{m s}^{-1}$

Open solution

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Q216. AIEEE 2012 (Online) 7th May

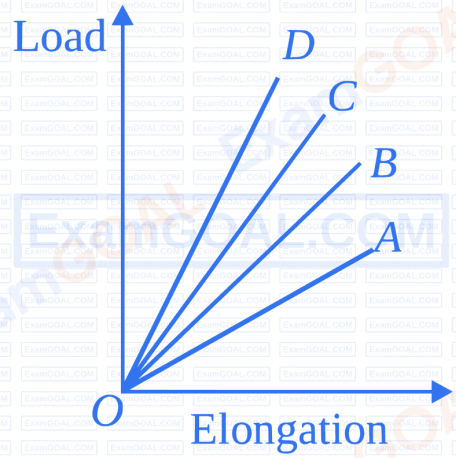
A structural steel rod has a radius of $10\ \text{mm}$ and length of $1.0\ \text{m}$. A $100\ \text{kN}$ force stretches it along its length. Young's modulus of structural steel is $2 \times 10^{11}\ \text{N m}^{-2}$. The percentage strain is about

(A) 0.16% (B) 0.32% (C) 0.08% (D) 0.24%

Open solution

Q217. AIEEE 2012 (Online) 7th May

The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line

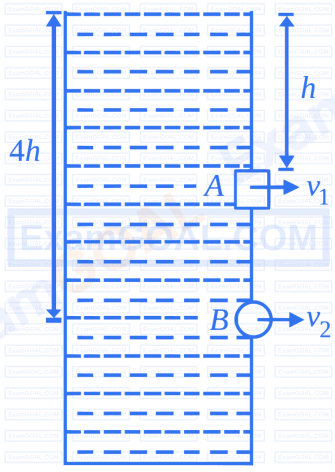


(A) OA (B) OC (C) OD (D) OB

Open solution

Q218. AIEEE 2012 (Online) 7th May

A square hole of side length ℓ is made at a depth of h and a circular hole of radius r is made at a depth of $4h$ from the surface of water in a water tank kept on a horizontal surface. If $\ell \ll h, r \ll h$ and the rate of water flow from the holes is the same, then r is equal to



(A) $\frac{\ell}{\sqrt{2\pi}}$ (B) $\frac{\ell}{\sqrt{3\pi}}$ (C) $\frac{\ell}{3\pi}$ (D) $\frac{\ell}{2\pi}$

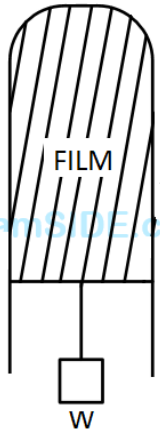
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Q219. AIEEE 2012

A thin liquid film formed between a U-shaped wire and a light slider supports a weight of $1.5 \times 10^{-2}\ N$ (see figure). The length of the slider is $30\ cm$ and its weight negligible. The surface tension of the liquid film is

Open solution



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(A) $0.0125\ Nm^{-1}$ (B) $0.1\ Nm^{-1}$ (C) $0.05\ Nm^{-1}$ (D) $0.025\ Nm^{-1}$

Q220. AIEEE 2011

Work done in increasing the size of a soap bubble from a radius of $3\ cm$ to $5\ cm$ is nearly (Surface tension of soap solution $= 0.03 Nm^{-1}$,

Open solution

(A) $0.2\pi mJ$ (B) $2\pi mJ$ (C) $0.4\pi mJ$ (D) $4\pi mJ$

Q221. AIEEE 2011

Water is flowing continuously from a tap having an internal diameter $8 \times 10^{-3}\ m$. The water velocity as it leaves the tap is $0.4\ ms^{-1}$. The diameter of the water stream at a distance $2 \times 10^{-1}\ m$ below the tap is close to :

Open solution

(A) $7.5 \times 10^{-3} m$ (B) $9.6 \times 10^{-3} m$ (C) $3.6 \times 10^{-3} m$ (D) $5.0 \times 10^{-3} m$

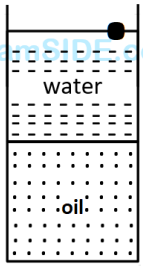
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Q222. AIEEE 2010

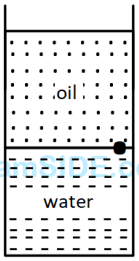
A ball is made of a material of density ρ where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?

Open solution

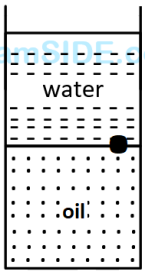
(A)



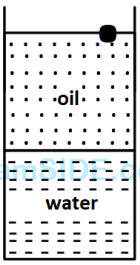
(B)



(C)



(D)



Q223. AIEEE 2009

Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

Open solution

(A) $4F$ (B) $6F$ (C) $9F$ (D) F

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Q224. AIEEE 2008

Open solution

A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density ρ_2 $(\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, i.e., $F_{viscous} = -kv^2$ $(k > 0)$. The terminal speed of the ball is

(A) $\sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

(B) $\frac{Vg\rho_1}{k}$

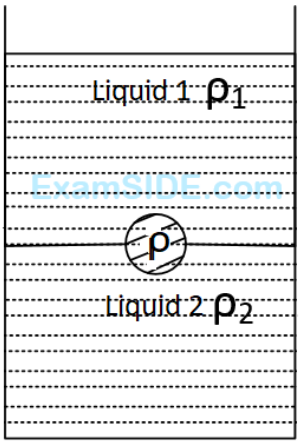
(C) $\sqrt{\frac{Vg\rho_1}{k}}$

(D) $\frac{Vg(\rho_1-\rho_2)}{k}$

Q225. AIEEE 2008

Open solution

A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_1 and ρ_3 ?



(A) $\rho_3 < \rho_1 < \rho_2$

(B) $\rho_1 > \rho_3 > \rho_2$

(C) $\rho_1 < \rho_2 < \rho_3$

(D) $\rho_1 < \rho_3 < \rho_2$

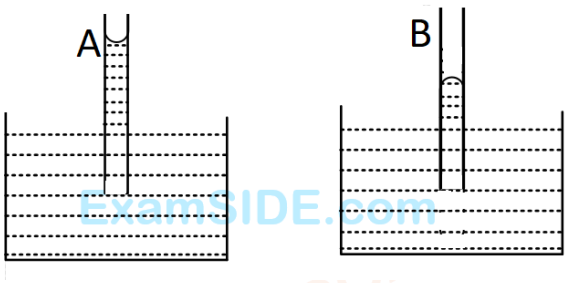
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Q226. AIEEE 2008

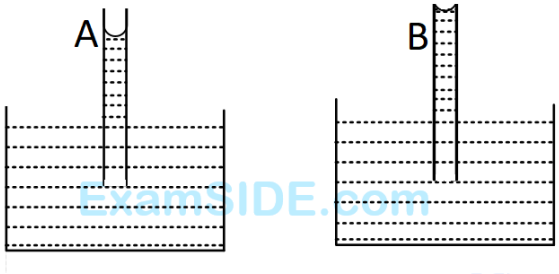
A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes?

Open solution

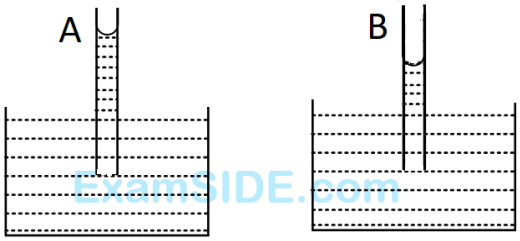
(A)



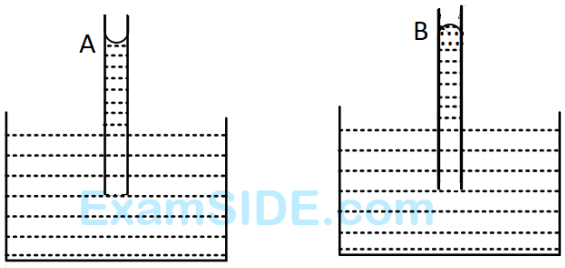
(B)



(C)



(D)



Q227. AIEEE 2006

A wire elongates by $l\ mm$ when a LOAD W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm)

(A) l (B) $2l$ (C) zero (D) $l/2$

Open solution

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Q228. AIEEE 2006

If the terminal speed of a sphere of gold (density $= 19.5\ kg/m^3$) is $0.2\ m/s$ in a viscous liquid (density $= 1.5\ kg/m^3$, find the terminal speed of a sphere of silver (density $= 10.5\ kg/m^3$) of the same size in the same liquid

Open solution

(A) $0.4\ m/s$ (B) $0.133\ m/s$ (C) $0.1\ m/s$ (D) $0.2\ m/s$

Q229. AIEEE 2005

If S is stress and Y is young's modulus of material of a wire, the energy stored in the wire per unit volume is

Open solution

(A) $\frac{S^2}{2Y}$ (B) $2S^2Y$ (C) $\frac{S}{2Y}$ (D) $\frac{2Y}{S^2}$

Q230. AIEEE 2005

A $20\ cm$ long capillary tube is dipped in water. The water rises up to $8\ cm$. If the entire arrangement is put in a freely falling elevator the length of water column in the capillary tube will be

Open solution

(A) $10\ cm$ (B) $8\ cm$ (C) $20\ cm$ (D) $4\ cm$

Q231. AIEEE 2004

A wire fixed at the upper end stretches by length l by applying a force F. The work done in stretching is

Open solution

(A) $2Fl$ (B) Fl (C) $\frac{F}{2l}$ (D) $\frac{Fl}{2}$

Q232. AIEEE 2004

Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v. The retarding viscous force acting on the spherical ball is

Open solution

(A) inversely proportional to both radius R and velocity v

(B) directly proportional to both radius R and velocity v

(C) directly proportional to R but inversely proportional to v

(D) inversely proportional to R, but directly proportional to velocity v

Q233. AIEEE 2004

If two soap bubbles of different radii are connected by a tube

Open solution

(A) air flows from the smaller bubble to the bigger

(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged

(C) air flows from the bigger bubble to the smaller bubble till the sizes become equal

(D) there is no flow of air.

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Q234. AIEEE 2003

A wire suspended vertically from one of its ends is stretched by attaching a weight of $200N$ to the lower end. The weight stretches the wire by $1\ mm$. Then the elastic energy stored in the wire is

(A) $0.2\ J$ (B) $10\ J$ (C) $20\ J$ (D) $0.1\ J$

Open solution

Q235. AIEEE 2002

A cylinder of height $20\ m$ is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom is

(A) 10 (B) 20 (C) 25.5 (D) 5

Open solution

Numerical 92 questions

Q236. JEE Main 2026 (Online) 6th April Morning Shift

A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{\text{rd}}$ of its initial volume ($V_o = 3$ litre) by applying required pressure. If the bulk modulus of the gas is $3 \times 10^5\ \text{N/m}^2$, the magnitude of work done on the gas is ____ J.

Open solution

Q237. JEE Main 2026 (Online) 5th April Evening Shift

A copper wire of length 3 m is stretched by 3 mm by applying an external force. The volume of the wire is $600 \times 10^{-6}\ \text{m}^3$. The elastic potential energy stored in the wire in stretched condition would be ____ J.

(Given Young modulus of copper $= 1.1 \times 10^{11}\ \text{N/m}^2$)

Open solution

Q238. JEE Main 2026 (Online) 5th April Morning Shift

A cube has side length 5 cm and modulus of rigidity $10^5\ \text{N/m}^2$. The displacement produced by a force of 10 N in the upper face of cube is ____ mm.

Open solution

Q239. JEE Main 2026 (Online) 4th April Morning Shift

The surface tension of a soap solution is $3.5 \times 10^{-2}\ \text{N/m}$. The work required to increase the radius of a soap bubble from 1 cm to 2 cm is $\alpha \times 10^{-6}$ J. The value of α is ____ .

($\pi = 22/7$)

Open solution

Q240. JEE Main 2026 (Online) 2nd April Morning Shift

A uniform wire of length l of weight w is suspended from the roof with a weight of W at the other end. The stress in the wire at $\frac{l}{3}$ distance from the top is $\left(\frac{W}{A} + \frac{2}{\gamma}\frac{w}{A}\right)$, where A is the cross sectional area of the wire. The value of γ is _____.

Open solution

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Q241. JEE Main 2026 (Online) 2nd April Morning Shift

A tub is filled with water and a wooden cube $10\text{ cm} \times 10\text{ cm} \times 10\text{ cm}$ is placed in the water. The wooden cube is found to float on the water with a part of it submerged in water. When a metal coin is placed on the wooden cube, the submerged part is increased by 3.87 cm. The mass of the metal coin is _____ gram.

(Take water density as 1 g/cm^3 and density of wood as 0.4 g/cm^3)

Open solution

Q242. JEE Main 2026 (Online) 24th January Evening Shift

A soap bubble of surface tension 0.04 N/m is blown to a diameter of 7 cm . If $(15000 - x)\mu\text{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is ____ .

$(\pi = 22/7)$

Open solution

Q243. JEE Main 2026 (Online) 24th January Morning Shift

Sixty four rain drops of radius 1 mm each falling down with a terminal velocity of 10 cm/s coalesce to form a bigger drop. The terminal velocity of bigger drop is

____ cm/s.

Open solution

Q244. JEE Main 2026 (Online) 21st January Evening Shift

The terminal velocity of a metallic ball of radius 6 mm in a viscous fluid is 20 cm/s. The terminal velocity of another ball of same material and having radius 3 mm in the same fluid will be ____ cm/s.

Open solution

Q245. JEE Main 2025 (Online) 8th April Evening Shift

A sample of a liquid is kept at 1 atm. It is compressed to 5 atm which leads to a change of volume of 0.8 cm^3. If the bulk modulus of the liquid is 2 GPa, the initial volume of the liquid was _____ litre.

(Take 1 atm = 10^5 Pa)

Open solution

Q246. JEE Main 2025 (Online) 8th April Evening Shift

A cube having a side of 10 cm with unknown mass and 200 gm mass were hung at two ends of a uniform rigid rod of 27 cm long. The rod along with masses was placed on a wedge keeping the distance between wedge point and 200 gm weight as 25 cm. Initially the masses were not at balance. A beaker is placed beneath the unknown mass and water is added slowly to it. At given point the masses were in balance and half volume of the unknown mass was inside the water.

(Take the density of unknown mass is more than that of the water, the mass did not absorb water and water density is 1 gm/cm^3.)

The unknown mass is ____ kg.

Open solution

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Q247. JEE Main 2025 (Online) 4th April Morning Shift

Two slabs with square cross section of different materials $(1, 2)$ with equal sides (l) and thickness d_1 and d_2 such that $d_2 = 2d_1$ and $l > d_2$. Considering lower edges of these slabs are fixed to the floor, we apply equal shearing force on the narrow faces. The angle of deformation is $\theta_2 = 2\theta_1$. If the shear moduli of material 1 is $4 \times 10^9\ \mathrm{N/m^2}$, then shear moduli of material 2 is $x \times 10^9\ \mathrm{N/m^2}$, where value of x is _____.

Open solution

Q248. JEE Main 2025 (Online) 3rd April Evening Shift

The excess pressure inside a soap bubble A in air is half the excess pressure inside another soap bubble B in air. If the volume of the bubble A is n times the volume of the bubble B, then, the value of n is_____.

Open solution

Q249. JEE Main 2025 (Online) 2nd April Evening Shift

The length of a light string is 1.4 m when the tension on it is 5 N . If the tension increases to 7 N , the length of the string is 1.56 m . The original length of the string is_____m.

Open solution

Q250. JEE Main 2025 (Online) 2nd April Morning Shift

A steel wire of length 2 m and Young's modulus $2.0 \times 10^{11}\ \mathrm{N\,m^{-2}}$ is stretched by a force. If Poisson ratio and transverse strain for the wire are 0.2 and 10^{-3} respectively, then the elastic potential energy density of the wire is _____ $\times 10^5$ (in SI units).

Open solution

Q251. JEE Main 2025 (Online) 2nd April Morning Shift

A vessel with square cross-section and height of 6 m is vertically partitioned. A small window of $100\ \mathrm{cm^2}$ with hinged door is fitted at a depth of 3 m in the partition wall. One part of the vessel is filled completely with water and the other side is filled with the liquid having density $1.5 \times 10^3\ \mathrm{kg/m^3}$. What force one needs to apply on the hinged door so that it does not get opened ?

(Acceleration due to gravity $= 10\ \mathrm{m/s^2}$)

Open solution

Q252. JEE Main 2025 (Online) 29th January Morning Shift

In a hydraulic lift, the surface area of the input piston is 6 cm^2 and that of the output piston is 1500 cm^2. If 100 N force is applied to the input piston to raise the output piston by 20 cm, then the work done is _____ kJ.

Open solution

Q253. JEE Main 2025 (Online) 28th January Evening Shift

The volume contraction of a solid copper cube of edge length 10 cm , when subjected to a hydraulic pressure of $7 \times 10^6\ \mathrm{Pa}$, would be _____ $\mathrm{mm^3}$.

(Given bulk modulus of copper $= 1.4 \times 10^{11}\ \mathrm{N\,m^{-2}}$)

Open solution

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Q254. JEE Main 2025 (Online) 24th January Evening Shift

The increase in pressure required to decrease the volume of a water sample by 0.2% is $\mathrm{P} \times 10^5 \mathrm{Nm}^{-2}$. Bulk modulus of water is $2.15 \times 10^9 \mathrm{Nm}^{-2}$. The value of P is _____ .

Open solution

Q255. JEE Main 2025 (Online) 23rd January Evening Shift

An air bubble of radius 1.0 mm is observed at a depth 20 cm below the free surface of a liquid having surface tension $0.095\ \mathrm{J/m^2}$ and density $10^3\ \mathrm{kg/m^3}$. The difference between pressure inside the bubble and atmospheric pressure is _____ $\mathrm{N/m^2}$. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Open solution

Q256. JEE Main 2025 (Online) 22nd January Morning Shift

Two soap bubbles of radius 2 cm and 4 cm , respectively, are in contact with each other. The radius of curvature of the common surface, in cm , is _____.

Open solution

Q257. JEE Main 2024 (Online) 9th April Morning Shift

Two persons pull a wire towards themselves. Each person exerts a force of $200\ \mathrm{N}$ on the wire. Young's modulus of the material of wire is $1 \times 10^{11}\ \mathrm{N\,m^{-2}}$. Original length of the wire is $2\ \mathrm{m}$ and the area of cross section is $2\ \mathrm{cm^2}$. The wire will extend in length by _____ $\mu\mathrm{m}$.

Open solution

Q258. JEE Main 2024 (Online) 8th April Evening Shift

Small water droplets of radius $0.01\ \mathrm{mm}$ are formed in the upper atmosphere and falling with a terminal velocity of $10\ \mathrm{cm/s}$. Due to condensation, if 8 such droplets are coalesced and formed a larger drop, the new terminal velocity will be _____ $\mathrm{cm/s}$.

Open solution

Q259. JEE Main 2024 (Online) 8th April Morning Shift

A liquid column of height $0.04\ \mathrm{cm}$ balances excess pressure of a soap bubble of certain radius. If density of liquid is $8 \times 10^3\ \mathrm{kg\,m^{-3}}$ and surface tension of soap solution is $0.28\ \mathrm{Nm^{-1}}$, then diameter of the soap bubble is _____ cm. (if $\mathrm{g} = 10\ \mathrm{m\,s^{-2}}$)

Open solution

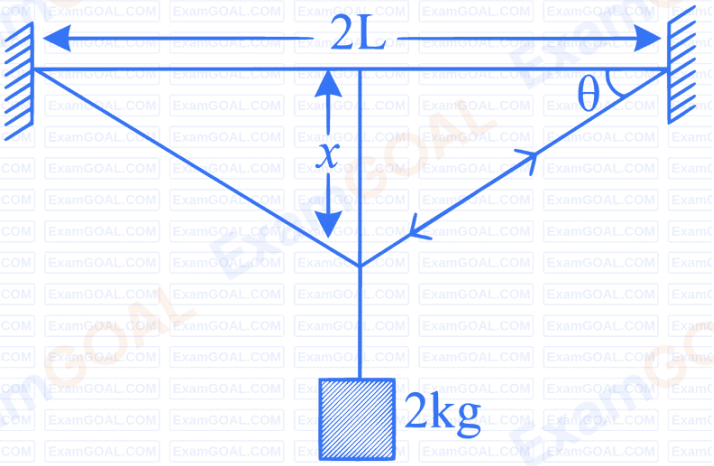
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Q260. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

A wire of cross sectional area A, modulus of elasticity $2 \times 10^{11}\ \mathrm{Nm}^{-2}$ and length $2\ \mathrm{m}$ is stretched between two vertical rigid supports. When a mass of $2\ \mathrm{kg}$ is suspended at the middle it sags lower from its original position making angle $\theta = \frac{1}{100}$ radian on the points of support. The value of A is _____ $\times 10^{-4}\ \mathrm{m}^2$ (consider $x << \mathrm{L}$).

(given : $\mathrm{g} = 10\ \mathrm{m/s}^2$)



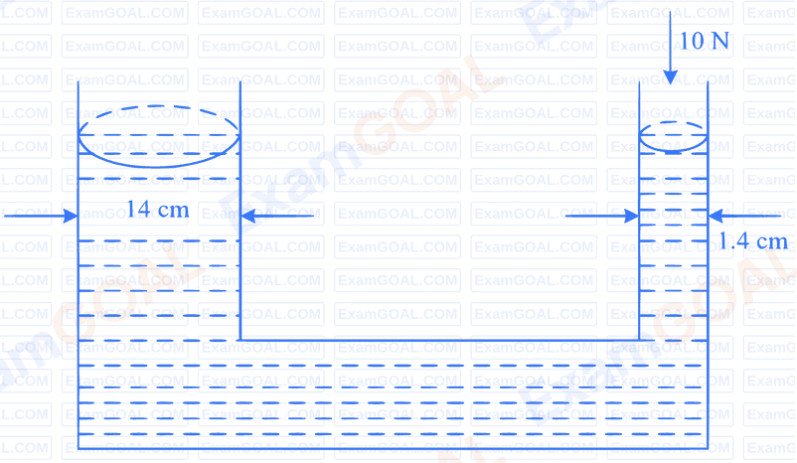
Q261. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

A big drop is formed by coalescing 1000 small droplets of water. The ratio of surface energy of 1000 droplets to that of energy of big drop is $\frac{10}{x}$. The value of x is _____.

Q262. JEE Main 2024 (Online) 5th April Evening Shift

Open solution



A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of $10\ \mathrm{N}$ is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.

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Q263. JEE Main 2024 (Online) 5th April Morning Shift

The density and breaking stress of a wire are $6 \times 10^4\ \mathrm{kg/m^3}$ and $1.2 \times 10^8\ \mathrm{N/m^2}$ respectively. The wire is suspended from a rigid support on a planet where acceleration due to gravity is $\frac{1}{3}^{\text{rd}}$ of the value on the surface of earth. The maximum length of the wire without breaking is _____ m (take, $g = 10\ \mathrm{m/s^2}$).

Open solution

Q264. JEE Main 2024 (Online) 4th April Evening Shift

Mercury is filled in a tube of radius $2\ \mathrm{cm}$ up to a height of $30\ \mathrm{cm}$. The force exerted by mercury on the bottom of the tube is _____ N.

(Given, atmospheric pressure $= 10^5\ \mathrm{Nm^{-2}}$, density of mercury $= 1.36 \times 10^4\ \mathrm{kg\,m^{-3}}$, $g = 10\ \mathrm{m\,s^{-2}}$, $\pi = \frac{22}{7}$)

Open solution

Q265. JEE Main 2024 (Online) 4th April Morning Shift

A soap bubble is blown to a diameter of $7\ \mathrm{cm}$. $36960\ \mathrm{erg}$ of work is done in blowing it further. If surface tension of soap solution is 40 dyne/cm then the new radius is _____ cm. Take $(\pi = \frac{22}{7})$.

Open solution

Q266. JEE Main 2024 (Online) 4th April Morning Shift

An elastic spring under tension of $3\ \mathrm{N}$ has a length a. Its length is b under tension $2\ \mathrm{N}$. For its length $(3a - 2b)$, the value of tension will be _____ N.

Open solution

Q267. JEE Main 2024 (Online) 1st February Evening Shift

One end of a metal wire is fixed to a ceiling and a load of $2\ \mathrm{kg}$ hangs from the other end. A similar wire is attached to the bottom of the load and another load of $1\ \mathrm{kg}$ hangs from this lower wire. Then the ratio of longitudinal strain of upper wire to that of the lower wire will be _____.

[Area of cross section of wire $= 0.005\ \mathrm{cm^2}$, $\mathrm{Y} = 2 \times 10^{11}\mathrm{Nm^{-2}}$ and $g = 10\ \mathrm{ms^{-2}}$]

Open solution

Q268. JEE Main 2024 (Online) 1st February Morning Shift

A plane is in level flight at constant speed and each of its two wings has an area of $40\ \mathrm{m^2}$. If the speed of the air is $180\ \mathrm{km/h}$ over the lower wing surface and $252\ \mathrm{km/h}$ over the upper wing surface, the mass of the plane is _____ kg.

(Take air density to be $1\ \mathrm{kg\,m^{-3}}$ and $g = 10\ \mathrm{ms^{-2}}$)

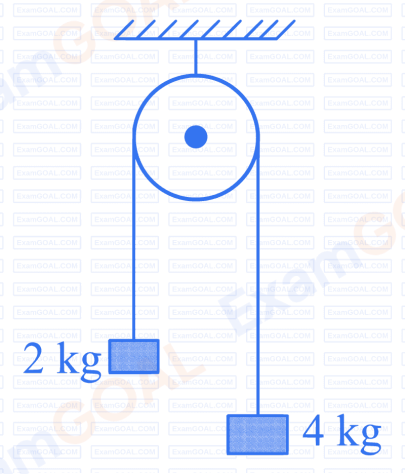
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Q269. JEE Main 2024 (Online) 31st January Evening Shift

Two blocks of mass $2\,\text{kg}$ and $4\,\text{kg}$ are connected by a metal wire going over a smooth pulley as shown in figure. The radius of wire is $4.0 \times 10^{-5}\,\text{m}$ and Young's modulus of the metal is $2.0 \times 10^{11}\,\text{N/m}^2$. The longitudinal strain developed in the wire is $\frac{1}{\alpha\pi}$. The value of α is _____. [Use $g = 10\,\text{m/s}^2$]

Open solution



Q270. JEE Main 2024 (Online) 31st January Morning Shift

The depth below the surface of sea to which a rubber ball be taken so as to decrease its volume by 0.02% is _____ m.

(Take density of sea water $= 10^3\text{kgm}^{-3}$, Bulk modulus of rubber $= 9 \times 10^8\,\text{Nm}^{-2}$, and $g = 10\,\text{ms}^{-2}$)

Open solution

Q271. JEE Main 2024 (Online) 30th January Evening Shift

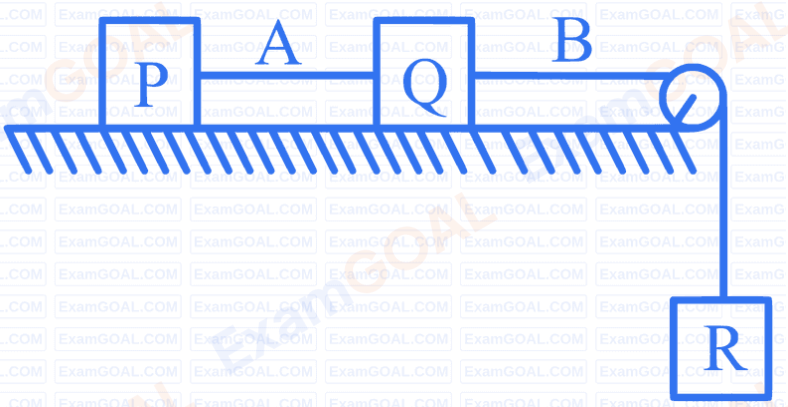
A big drop is formed by coalescing 1000 small identical drops of water. If E_1 be the total surface energy of 1000 small drops of water and E_2 be the surface energy of single big drop of water, then $E_1 : E_2$ is $x : 1$ where $x =$ _____.

Open solution

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Q272. JEE Main 2024 (Online) 30th January Morning Shift

Each of three blocks P, Q and R shown in figure has a mass of $3\ \mathrm{kg}$. Each of the wires A and B has cross-sectional area $0.005\ \mathrm{cm}^2$ and Young's modulus $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$. Neglecting friction, the longitudinal strain on wire B is _____ $\times 10^{-4}$. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



Open solution

Q273. JEE Main 2024 (Online) 29th January Evening Shift

Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio $4 : 1$ and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2.

The value of $\frac{F_1}{F_2}$ is _____.

Open solution

Q274. JEE Main 2024 (Online) 29th January Morning Shift

In a test experiment on a model aeroplane in wind tunnel, the flow speeds on the upper and lower surfaces of the wings are $70\ \mathrm{ms^{-1}}$ and $65\ \mathrm{ms^{-1}}$ respectively. If the wing area is $2\ \mathrm{m^2}$, the lift of the wing is _____ N.

(Given density of air $= 1.2\ \mathrm{kg\,m^{-3}}$)

Open solution

Q275. JEE Main 2024 (Online) 27th January Evening Shift

The reading of pressure metre attached with a closed pipe is $4.5 \times 10^4\ \mathrm{N/m^2}$. On opening the valve, water starts flowing and the reading of pressure metre falls to $2.0 \times 10^4\ \mathrm{N/m^2}$. The velocity of water is found to be $\sqrt{V}\ \mathrm{m/s}$. The value of V is _____.

Open solution

Q276. JEE Main 2024 (Online) 27th January Morning Shift

If average depth of an ocean is $4000\ \mathrm{m}$ and the bulk modulus of water is $2 \times 10^9\ \mathrm{Nm^{-2}}$, then fractional compression $\frac{\Delta V}{V}$ of water at the bottom of ocean is $\alpha \times 10^{-2}$. The value of α is _____

(Given, $\mathrm{g} = 10\ \mathrm{ms^{-2}}$, $\rho = 1000\ \mathrm{kg\,m^{-3}}$)

Open solution

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Q277. JEE Main 2023 (Online) 15th April Morning Shift

There is an air bubble of radius $1.0\ \text{mm}$ in a liquid of surface tension $0.075\ \text{Nm}^{-1}$ and density $1000\ \text{kg m}^{-3}$ at a depth of $10\ \text{cm}$ below the free surface. The amount by which the pressure inside the bubble is greater than the atmospheric pressure is _____ $\text{Pa}\left(\text{g} = 10\ \text{ms}^{-2}\right)$

Open solution

Q278. JEE Main 2023 (Online) 13th April Morning Shift

The elastic potential energy stored in a steel wire of length $20\ \text{m}$ stretched through $2\ \text{cm}$ is $80\ \text{J}$. The cross sectional area of the wire is _____ mm^2.

(Given, $Y = 2.0 \times 10^{11}\text{Nm}^{-2}$)

Open solution

Q279. JEE Main 2023 (Online) 12th April Morning Shift

Glycerin of density $1.25 \times 10^3\ \text{kg m}^{-3}$ is flowing through the conical section of pipe. The area of cross-section of the pipe at its ends are $10\ \text{cm}^2$ and $5\ \text{cm}^2$ and pressure drop across its length is $3\ \text{Nm}^{-2}$. The rate of flow of glycerin through the pipe is $x \times 10^{-5}\ \text{m}^3\ \text{s}^{-1}$. The value of x is _____.

Open solution

Q280. JEE Main 2023 (Online) 11th April Evening Shift

The surface tension of soap solution is $3.5 \times 10^{-2}\ \text{Nm}^{-1}$. The amount of work done required to increase the radius of soap bubble from $10\ \text{cm}$ to $20\ \text{cm}$ is _____ $\times 10^{-4}\ \text{J}$.

$(\text{take}\ \pi = 22/7)$

Open solution

Q281. JEE Main 2023 (Online) 11th April Morning Shift

The length of a wire becomes l_1 and l_2 when $100\ \text{N}$ and $120\ \text{N}$ tensions are applied respectively. If $10\,l_2 = 11\,l_1$, the natural length of wire will be $\frac{1}{x}\,l_1$. Here the value of x is _____.

Open solution

Q282. JEE Main 2023 (Online) 10th April Evening Shift

Figure below shows a liquid being pushed out of the tube by a piston having area of cross section $2.0\ \text{cm}^2$. The area of cross section at the outlet is $10\ \text{mm}^2$. If the piston is pushed at a speed of $4\ \text{cm s}^{-1}$, the speed of outgoing fluid is _____ cms^{-1}

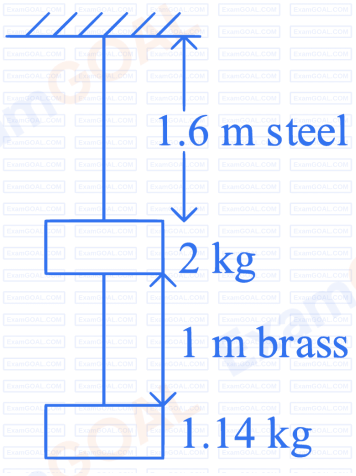


Open solution

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Q283. JEE Main 2023 (Online) 10th April Morning Shift

Two wires each of radius 0.2 cm and negligible mass, one made of steel and the other made of brass are loaded as shown in the figure. The elongation of the steel wire is _____ $\times 10^{-6}$ m. [Young's modulus for steel = 2×10^{11} Nm^{-2} and g = 10 ms^{-2}]



Open solution

Q284. JEE Main 2023 (Online) 8th April Morning Shift

An air bubble of diameter $6\ \mathrm{mm}$ rises steadily through a solution of density $1750\ \mathrm{kg/m^3}$ at the rate of $0.35\ \mathrm{cm/s}$. The co-efficient of viscosity of the solution (neglect density of air) is _____ Pas (given, $\mathrm{g} = 10\ \mathrm{ms}^{-2}$).

Open solution

Q285. JEE Main 2023 (Online) 6th April Evening Shift

A metal block of mass m is suspended from a rigid support through a metal wire of diameter $14\ \mathrm{mm}$. The tensile stress developed in the wire under equilibrium state is $7 \times 10^5 \mathrm{Nm}^{-2}$. The value of mass m is _____ kg. (Take, $\mathrm{g} = 9.8\ \mathrm{ms}^{-2}$ and $\pi = \frac{22}{7}$)

Open solution

Q286. JEE Main 2023 (Online) 6th April Morning Shift

A steel rod has a radius of $20\ \mathrm{mm}$ and a length of $2.0\ \mathrm{m}$. A force of $62.8\ \mathrm{kN}$ stretches it along its length. Young's modulus of steel is $2.0 \times 10^{11}\ \mathrm{N/m^2}$. The longitudinal strain produced in the wire is _____ $\times 10^{-5}$

Open solution

Q287. JEE Main 2023 (Online) 1st February Evening Shift

The surface of water in a water tank of cross section area $750\ \mathrm{cm}^2$ on the top of a house is $h\ \mathrm{m}$ above the tap level. The speed of water coming out through the tap of cross section area $500\ \mathrm{mm}^2$ is $30\ \mathrm{cm/s}$. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3}\ \mathrm{m/s}$. The value of x will be _____.

Open solution

Q288. JEE Main 2023 (Online) 1st February Morning Shift

A certain pressure 'P' is applied to 1 litre of water and 2 litre of a liquid separately. Water gets compressed to 0.01% whereas the liquid gets compressed to 0.03%. The ratio of Bulk modulus of water to that of the liquid is $\frac{3}{x}$. The value of x is _____.

Open solution

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Q289. JEE Main 2023 (Online) 31st January Morning Shift

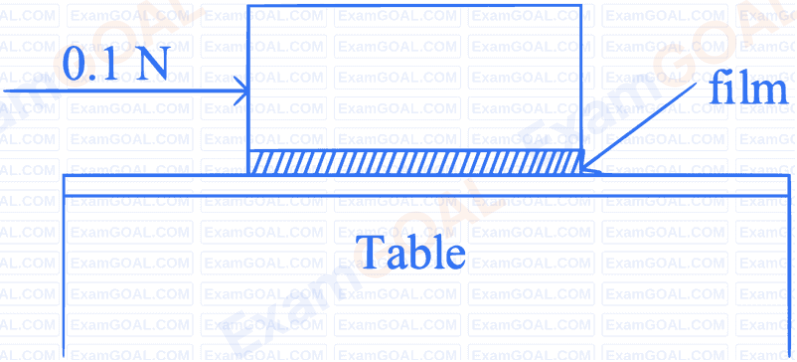
A thin rod having a length of $1\ \mathrm{m}$ and area of cross-section $3 \times 10^{-6}\ \mathrm{m}^2$ is suspended vertically from one end. The rod is cooled from $210^{\circ}\mathrm{C}$ to $160^{\circ}\mathrm{C}$. After cooling, a mass M is attached at the lower end of the rod such that the length of rod again becomes $1\ \mathrm{m}$. Young's modulus and coefficient of linear expansion of the rod are $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$ and $2 \times 10^{-5}\ \mathrm{K^{-1}}$, respectively. The value of M is _____ kg.

(Take $\mathrm{g} = 10\ \mathrm{m\,s^{-2}}$)

Open solution

Q290. JEE Main 2023 (Online) 29th January Evening Shift

A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is 5.0×10^{-3} Pl, the speed of block is _____ $\times 10^{-3}$ m/s.



Open solution

Q291. JEE Main 2023 (Online) 25th January Evening Shift

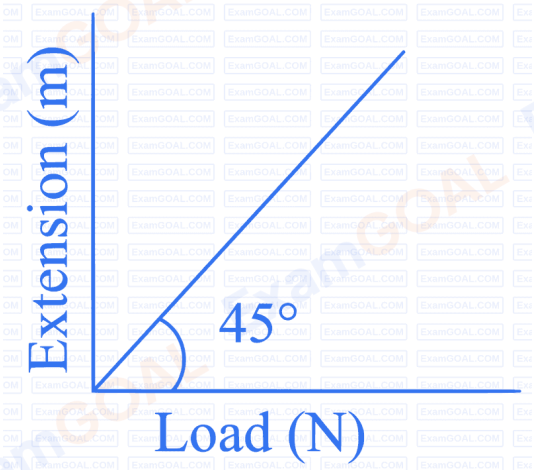
A spherical drop of liquid splits into 1000 identical spherical drops. If u_i is the surface energy of the original drop and u_f is the total surface energy of the resulting drops, then (ignoring evaporation), $\frac{u_f}{u_i} = \left(\frac{10}{x}\right)$. Then value of x is _____ :

Open solution

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Q292. JEE Main 2023 (Online) 25th January Morning Shift

As shown in the figure, in an experiment to determine Young's modulus of a wire, the extension-load curve is plotted. The curve is a straight line passing through the origin and makes an angle of 45° with the load axis. The length of wire is 62.8 cm and its diameter is 4 mm. The Young's modulus is found to be $x \times 10^4$ Nm^{-2}. The value of x is _____.



Open solution

Q293. JEE Main 2023 (Online) 24th January Evening Shift

A Spherical ball of radius 1mm and density 10.5 g/cc is dropped in glycerine of coefficient of viscosity 9.8 poise and density 1.5 g/cc. Viscous force on the ball when it attains constant velocity is 3696×10^{-x} N. The value of x is _____.

(Given, g = 9.8 m/s^2 and $\pi = \frac{22}{7}$)

Open solution

Q294. JEE Main 2022 (Online) 30th July Evening Shift

A water drop of radius $1\ \text{cm}$ is broken into eight equal droplets. Surface tension of water is $0.075\ \text{N m}^{-1}$. The gain in surface energy is _____ $\times 10^{-7}$ J. (Take $\pi = 3.14$)

Open solution

Q295. JEE Main 2022 (Online) 29th July Evening Shift

A tube of length $50\ \text{cm}$ is filled completely with an incompressible liquid of mass $250\ \text{g}$ and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}\ \text{rad s}^{-1}$. If F be the force exerted by the liquid at the other end then the value of x will be _____.

Open solution

Q296. JEE Main 2022 (Online) 29th July Evening Shift

A metal wire of length $0.5\ \text{m}$ and cross-sectional area $10^{-4}\ \text{m}^2$ has breaking stress $5 \times 10^8\ \text{Nm}^{-2}$. A block of $10\ \text{kg}$ is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be _____ ms^{-1}.

Open solution

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Q297. JEE Main 2022 (Online) 29th July Evening Shift

The velocity of a small ball of mass $0.3\ \mathrm{g}$ and density $8\ \mathrm{g/cc}$ when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $1.3\ \mathrm{g/cc}$, then the value of viscous force acting on the ball will be $x \times 10^{-4}\ \mathrm{N}$. The value of x is _____. [use $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Open solution

Q298. JEE Main 2022 (Online) 29th July Evening Shift

The speed of a transverse wave passing through a string of length $50\ \mathrm{cm}$ and mass $10\ \mathrm{g}$ is $60\ \mathrm{ms^{-1}}$. The area of cross-section of the wire is $2.0\ \mathrm{mm^2}$ and its Young's modulus is $1.2 \times 10^{11}\mathrm{Nm^{-2}}$. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}\ \mathrm{m}$. The value of x is _____.

Open solution

Q299. JEE Main 2022 (Online) 28th July Evening Shift

A string of area of cross-section $4\ \mathrm{mm^2}$ and length $0.5\ \mathrm{m}$ is connected with a rigid body of mass $2\ \mathrm{kg}$. The body is rotated in a vertical circular path of radius $0.5\ \mathrm{m}$. The body acquires a speed of $5\ \mathrm{m/s}$ at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is _____ $\times 10^{-5}$.

(use Young's modulus $10^{11}\ \mathrm{N/m^2}$ and $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Open solution

Q300. JEE Main 2022 (Online) 28th July Morning Shift

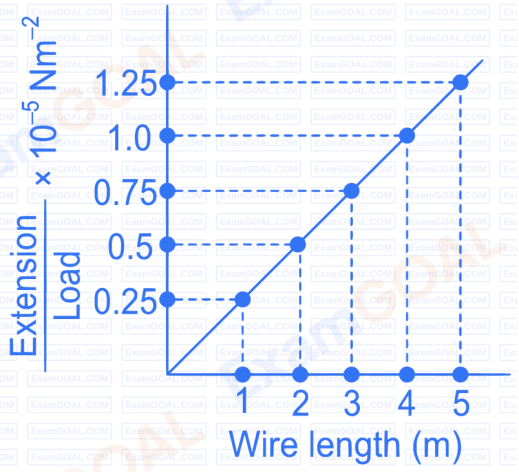
The diameter of an air bubble which was initially $2\ \mathrm{mm}$, rises steadily through a solution of density $1750\ \mathrm{kg\ m^{-3}}$ at the rate of $0.35\ \mathrm{cms^{-1}}$. The coefficient of viscosity of the solution is _____ poise (in nearest integer). (the density of air is negligible).

Open solution

Q301. JEE Main 2022 (Online) 27th July Evening Shift

In an experiment to determine the Young's modulus, steel wires of five different lengths $(1, 2, 3, 4,$ and $5\ \mathrm{m})$ but of same cross section $\left(2\ \mathrm{mm^2}\right)$ were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}\ \mathrm{Nm^{-2}}$, then the value of x is _____.

Open solution



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Q302. JEE Main 2022 (Online) 27th July Evening Shift

A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of another single soap bubble of radius r cm. The value of r is _____.

Open solution

Q303. JEE Main 2022 (Online) 27th July Morning Shift

A square aluminum (shear modulus is $25 \times 10^9\ \mathrm{Nm}^{-2}$) slab of side $60\ \mathrm{cm}$ and thickness $15\ \mathrm{cm}$ is subjected to a shearing force (on its narrow face) of $18.0 \times 10^4\ \mathrm{N}$. The lower edge is riveted to the floor. The displacement of the upper edge is _____ μm.

Open solution

Q304. JEE Main 2022 (Online) 26th July Evening Shift

A uniform heavy rod of mass $20\ \mathrm{kg}$, cross sectional area $0.4\ \mathrm{m}^2$ and length $20\ \mathrm{m}$ is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9}\ \mathrm{m}$. The value of x is _____.

(Given, young modulus Y = 2×10^{11} Nm^{-2} and g = 10 ms^{-2})

Open solution

Q305. JEE Main 2022 (Online) 26th July Morning Shift

In an experiment to determine the Young's modulus of wire of a length exactly $1\ \mathrm{m}$, the extension in the length of the wire is measured as $0.4\ \mathrm{mm}$ with an uncertainty of $\pm 0.02\ \mathrm{mm}$ when a load of $1\ \mathrm{kg}$ is applied. The diameter of the wire is measured as $0.4\ \mathrm{mm}$ with an uncertainty of $\pm 0.01\ \mathrm{mm}$. The error in the measurement of Young's modulus $(\Delta \mathrm{Y})$ is found to be $x \times 10^{10}\ \mathrm{Nm}^{-2}$. The value of x is _____. (take $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

Open solution

Q306. JEE Main 2022 (Online) 25th July Morning Shift

A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F, its length increases by $5\ \mathrm{cm}$. Another wire of the same material of length $4\mathrm{L}$ and radius $4\mathrm{r}$ is pulled by a force $4\mathrm{F}$ under same conditions. The increase in length of this wire is _____ cm.

Open solution

Q307. JEE Main 2022 (Online) 30th June Morning Shift

The excess pressure inside a liquid drop is 500 Nm^{-2}. If the radius of the drop is 2 mm, the surface tension of liquid is x $\times$ 10^{-3} Nm^{-1}. The value of x is _____.

Open solution

Q308. JEE Main 2022 (Online) 29th June Evening Shift

A small spherical ball of radius 0.1 mm and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If, after entering the water the velocity of ball does not change and it continue to fall with same constant velocity inside water, then the value of h will be _____ m.

(Given g = 10 ms^{-2}, viscosity of water = 1.0×10^{-5} N-sm^{-2}).

Open solution

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Q309. JEE Main 2022 (Online) 28th June Evening Shift

A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from $A_1 = 1.2 \times 10^{-2}\ m^2$ to $A_2 = \frac{A_1}{2}$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____ $\times 10^{-3}\ m^3s^{-1}$.

Open solution

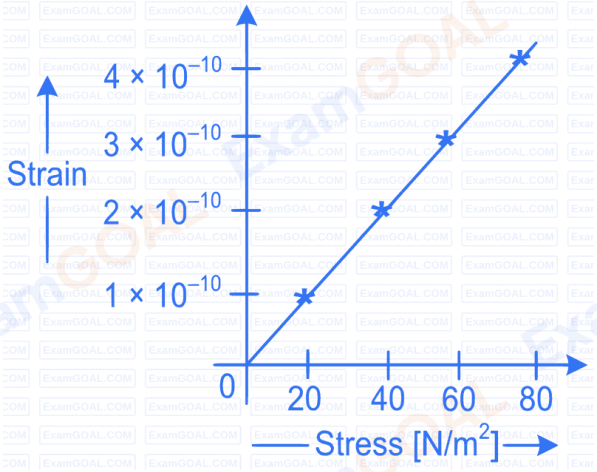
Q310. JEE Main 2022 (Online) 27th June Morning Shift

The area of cross-section of a large tank is 0.5 m^2. It has a narrow opening near the bottom having area of cross-section 1 cm^2. A load of 25 kg is applied on the water at the top in the tank. Neglecting the speed of water in the tank, the velocity of the water, coming out of the opening at the time when the height of water level in the tank is 40 cm above the bottom, will be _____ cms $^{-1}$. [Take g = 10 ms^{-2}]

Open solution

Q311. JEE Main 2022 (Online) 26th June Morning Shift

The elastic behaviour of material for linear stress and linear strain, is shown in the figure. The energy density for a linear strain of 5×10^{-4} is _____ kJ/m^3. Assume that material is elastic upto the linear strain of 5×10^{-4}.



Open solution

Q312. JEE Main 2022 (Online) 26th June Morning Shift

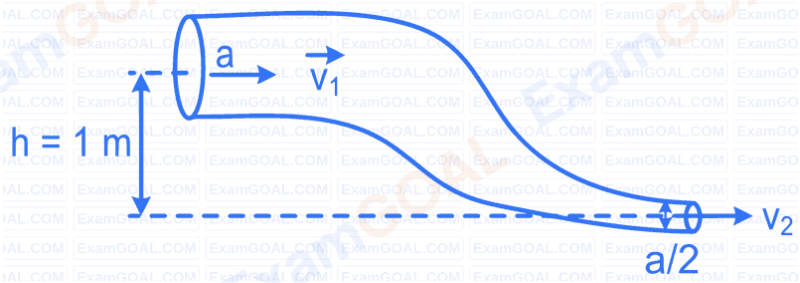
The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5} m on another planet. The acceleration due to gravity on the planet will be _____ ms^{-2}. (Take acceleration due to gravity on the surface of earth = 10 ms^{-2})

Open solution

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Q313. JEE Main 2022 (Online) 26th June Morning Shift

An ideal fluid of density 800 kgm^{-3}, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6}$ ms^{-1} for x = _____. (Given g = 10 ms^{-2})



Open solution

Q314. JEE Main 2022 (Online) 25th June Morning Shift

The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is _____ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)

Open solution

Q315. JEE Main 2021 (Online) 1st September Evening Shift

A steel rod with y = 2.0×10^{11} Nm^{-2} and $\alpha = 10^{-5}\,^{\circ}C^{-1}$ of length 4 m and area of cross-section 10 cm^2 is heated from 0°C to 400°C without being allowed to extend. The tension produced in the rod is $x \times 10^5$ N where the value of x is _____.

Open solution

Q316. JEE Main 2021 (Online) 31st August Morning Shift

When a rubber ball is taken to a depth of _____m in deep sea, its volume decreases by 0.5%.

(The bulk modulus of rubber = 9.8×10^8 Nm^{-2}, Density of sea water = 10^3 kgm^{-3}, g = 9.8 m/s^2)

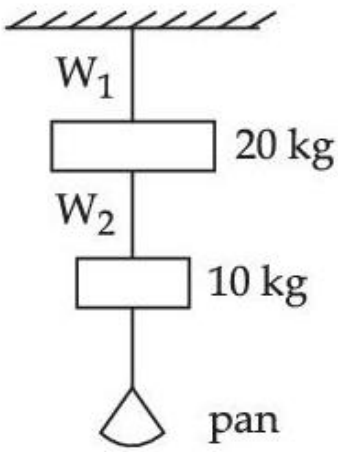
Open solution

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Q317. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

Wires W_1 and W_2 are made of same material having the breaking stress of 1.25×10^9 N/m^2. W_1 and W_2 have cross-sectional area of 8×10^{-7} m^2 and 4×10^{-7} m^2, respectively. Masses of 20 kg and 10 kg hang from them as shown in the figure. The maximum mass that can be placed in the pan without breaking the wires is _____ kg. (Use g = 10 m/s^2)



Q318. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is ..... cm.

Q319. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

The water is filled upto height of 12 m in a tank having vertical sidewalls. A hole is made in one of the walls at a depth 'h' below the water level. The value of 'h' for which the emerging steam of water strikes the ground at the maximum range is _____ m.

Q320. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

A stone of mass 20 g is projected from a rubber catapult of length 0.1 m and area of cross section 10^{-6} m^2 stretched by an amount 0.04 m. The velocity of the projected stone is _____ m/s.

(Young's modulus of rubber = 0.5×10^9 N/m^2)

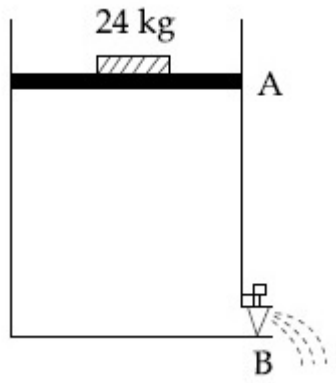
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Q321. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Consider a water tank as shown in the figure. It's cross-sectional area is 0.4 m^2. The tank has an opening B near the bottom whose cross-section area is 1 cm^2. A load of 24 kg is applied on the water at the top when the height of the water level is 40 cm above the bottom, the velocity of water coming out the opening B is v ms^{-1}.

The value of v, to the nearest integer, is _____. [Take value of g to be 10 ms^{-2}]



Q322. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

Two separate wires A and B are stretched by 2 mm and 4 mm respectively, when they are subjected to a force of 2 N. Assume that both the wires are made up of same material and the radius of wire B is 4 times that of the radius of wire A. The length of the wires A and B are in the ratio of a : b. Then a/b can be expressed as 1/x where x is _____.

Q323. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3}$ cm. Then the thickness of oleic acid layer will be x $\times 10^{-14}$ m. Where x is _____.

Q324. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F. The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

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Q325. JEE Main 2021 (Online) 24th February Morning Shift

A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

Open solution

Q326. JEE Main 2020 (Online) 3rd September Morning Slot

When a long glass capillary tube of radius 0.015 cm is dipped in a liquid, the liquid rises to a height of 15 cm within it. If the contact angle between the liquid and glass to close to 0°, the surface tension of the liquid, in milliNewton m–1, is [$\rho_{(liquid)}$ = 900 kgm^{-3}, g = 10 ms^{-2}]
(Give answer in closest integer) ____.

Open solution

Q327. JEE Main 2020 (Online) 9th January Morning Slot

A body of mass m = 10 kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is :
(Breaking stress of wire = 4.8×10^7 Nm^{-2} and
area of cross-section of the wire = 10^{-2} cm^2) is:

Open solution

Out of syllabus 15 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q328. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

A body cools from $80^\circ C$ to $60^\circ C$ in 5 minutes. The temperature of the surrounding is $20^\circ C$. The time it takes to cool from $60^\circ C$ to $40^\circ C$ is :

(A) 500 s (B) $\frac{25}{3}$ s (C) 450 s (D) 420 s

Open solution

Q329. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

A body cools in 7 minutes from $60^\circ C$ to $40^\circ C$. The temperature of the surrounding is $10^\circ C$. The temperature of the body after the next 7 minutes will be:

(A) $34^\circ C$ (B) $28^\circ C$ (C) $32^\circ C$ (D) $30^\circ C$

Open solution

Q330. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

A bowl filled with very hot soup cools from 98°C to 86°C in 2 minutes when the room temperature is 22°C. How long it will take to cool from 75°C to 69°C?

(A) 2 minutes (B) 0.5 minute (C) 1.4 minutes (D) 1 minute

Open solution

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Q331. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

A body takes 4 min. to cool from 61° C to 59° C. If the temperature of the surroundings is 30° C, the time taken by the body to cool from 51° C to 49° C is :

(A) 4 min. (B) 3 min. (C) 8 min. (D) 6 min.

Open solution

Q332. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

A metallic sphere cools from 50°C to 40° in 300 s. If atmospheric temperature around is 20°C, then the sphere's temperature after the next 5 minutes will be close to :

(A) 35°C (B) 31°C (C) 33°C (D) 28°C

Open solution

Q333. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A body takes 10 minutes to cool from 60°C to 50°C. The tempertature of surroundings is constant at 25°C. Then, the temperature of the body after next 10 minutes will be approximately :

(A) 47°C (B) 41°C (C) 45°C (D) 43°C

Open solution

Q334. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Hot water cools from 60°C to 50°C in the first 10 minutes and 42°C in the next 10 minutes. The temperature of the surroundings is :

(A) 25°C (B) 10°C (C) 15°C (D) 20°C

Open solution

Q335. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A hot body, obeying Newton's law of cooling is cooling down from its peak value 80°C to an ambient temperature of 30°C. It takes 5 minutes in cooling down from 80°C to 40°C. How much time will it take to cool down from 62°C to 32°C ? (Given ℓn 2 = 0.693, ℓn 5 = 1.609)

(A) 3.75 minutes (B) 8.6 minutes (C) 9.6 minutes (D) 6.5 minutes

Open solution

Q336. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

A mass of 50 g of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from 30°C to 25°C. A mass of 100 g of another liquid in an identical vessel with identical surroundings takes the same time to cool from 30°C to 25°C. The specific heat of the liquid is : (The water equivalent of the vessel is 30 g.)

(A) 2.0 kcal/kg (B) 7 kcal/kg (C) 3 kcal/kg (D) 0.5 kcal/kg

Open solution

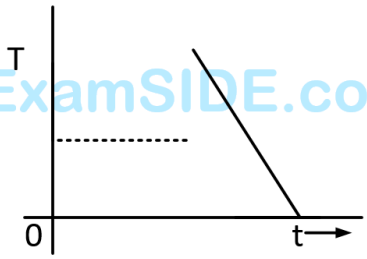
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Q337. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

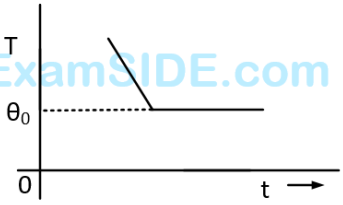
Open solution

If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0, the graph between the temperature T of the metal and time t will be closest to

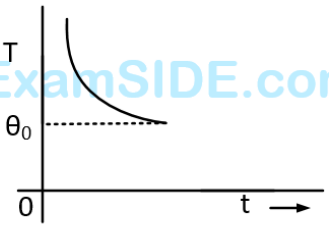
(A)



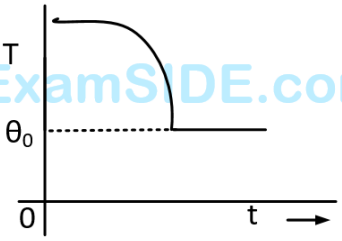
(B)



(C)



(D)



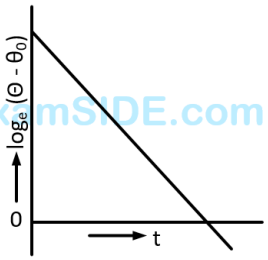
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Q338. AIEEE 2012 · MCQ (Single Correct Answer)

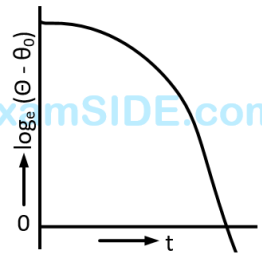
A liquid in a beaker has temperature $\theta(t)$ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e(\theta - \theta_0)$ and t is :

Open solution

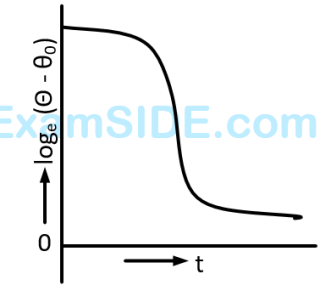
(A)



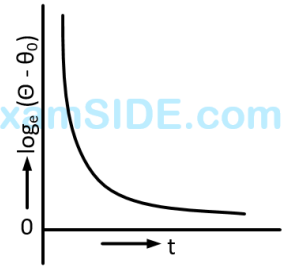
(B)



(C)



(D)



Q339. AIEEE 2003 · MCQ (Single Correct Answer)

According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surrounding, and n is equal to :

(A) two (B) three (C) four (D) one

Open solution

Q340. JEE Main 2023 (Online) 29th January Morning Shift · Numerical

A body cools from 60°C to 40°C in 6 minutes. If, temperature of surroundings is 10°C. Then, after the next 6 minutes, its temperature will be _____ °C.

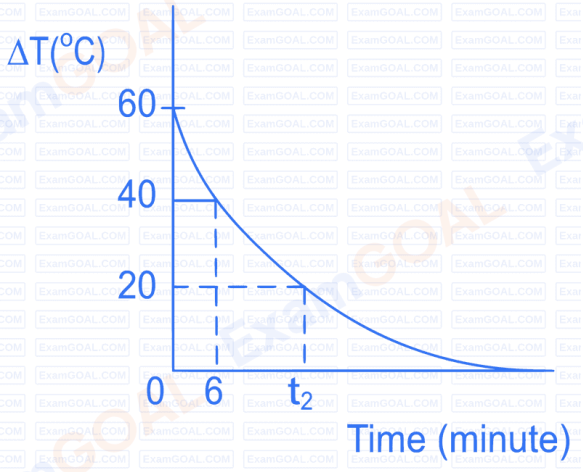
Open solution

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Q341. JEE Main 2022 (Online) 24th June Evening Shift · **Numerical**

In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C. The value of t_2 as mentioned in the graph will be _____.

Open solution



Q342. JEE Main 2021 (Online) 22th July Evening Shift · **Numerical**

In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is _____° C.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **D**	3. **C**	4. **B**	5. **A**
6. **C**	7. **B**	8. **C**	9. **B**	10. **D**
11. **B**	12. **B**	13. **B**	14. **C**	15. **B**
16. **B**	17. **D**	18. **D**	19. **A**	20. **D**
21. **D**	22. **B**	23. **A**	24. **B**	25. **D**
26. **D**	27. **A**	28. **B**	29. **A**	30. **D**
31. **C**	32. **D**	33. **D**	34. **C**	35. **B**
36. **B**	37. **C**	38. **B**	39. **B**	40. **D**
41. **B**	42. **D**	43. **D**	44. **C**	45. **B**
46. **A**	47. **D**	48. **D**	49. **A**	50. **D**
51. **B**	52. **A**	53. **B**	54. **B**	55. **D**
56. **A**	57. **B**	58. **C**	59. **D**	60. **B**
61. **D**	62. **B**	63. **A**	64. **C**	65. **C**
66. **B**	67. **C**	68. **B**	69. **B**	70. **C**
71. **A**	72. **C**	73. **D**	74. **D**	75. **D**
76. **A**	77. **D**	78. **A**	79. **C**	80. **D**
81. **B**	82. **C**	83. **A**	84. **A**	85. **A**
86. **B**	87. **D**	88. **A**	89. **A**	90. **B**
91. **B**	92. **D**	93. **A**	94. **A**	95. **D**
96. **C**	97. **B**	98. **C**	99. **D**	100. **B**
101. **A**	102. **C**	103. **A**	104. **B**	105. **C**
106. **C**	107. **D**	108. **C**	109. **A**	110. **B**
111. **B**	112. **C**	113. **C**	114. **C**	115. **B**
116. **D**	117. **B**	118. **C**	119. **B**	120. **C**
121. **C**	122. **C**	123. **B**	124. **B**	125. **A**
126. **C**	127. **B**	128. **D**	129. **A**	130. **B**
131. **D**	132. **B**	133. **C**	134. **D**	135. **B**
136. **A**	137. **B**	138. **D**	139. **C**	140. **C**
141. **D**	142. **B**	143. **B**	144. **C**	145. **D**
146. **D**	147. **D**	148. **C**	149. **B**	150. **C**

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151. **B**	152. **D**	153. **A**	154. **A**	155. **C**
156. **C**	157. **A**	158. **B**	159. **A**	160. **A**
161. **C**	162. **D**	163. **A**	164. **B**	165. **B**
166. **A**	167. **C**	168. **B**	169. **A**	170. **A**
171. **B**	172. **B**	173. **B**	174. **D**	175. **C**
176. **A**	177. **C**	178. **D**	179. **B**	180. **C**
181. **D**	182. **B**	183. **B**	184. **C**	185. **A**
186. *Scan QR*	187. **C**	188. **A**	189. **D**	190. **C**
191. **B**	192. **B**	193. **A**	194. **B**	195. **A**
196. **C**	197. **A**	198. **A**	199. **A**	200. **D**
201. **C**	202. **C**	203. **D**	204. **C**	205. **C**
206. **C**	207. **C**	208. **B**	209. **C**	210. **D**
211. **A**	212. **B**	213. **C**	214. **B**	215. **A**
216. **A**	217. **A**	218. **A**	219. **D**	220. **C**
221. **C**	222. **B**	223. **C**	224. **A**	225. **D**
226. **C**	227. **A**	228. **C**	229. **A**	230. **C**
231. **D**	232. **B**	233. **A**	234. **D**	235. **B**

Numerical

236. **989**	237. **33**	238. **2**	239. **264**	240. **3**
241. **387**	242. **11304**	243. **160**	244. **5**	245. **4**
246. **3**	247. **1**	248. **8**	249. **1**	250. **25**
251. **150**	252. **5**	253. **50**	254. **43**	255. **2190**
256. **4**	257. **20**	258. **40**	259. **7**	260. **1**
261. **1**	262. **1000**	263. **600**	264. **177**	265. **7**
266. **5**	267. **3**	268. **9600**	269. **12**	270. **18**
271. **10**	272. **2**	273. **16**	274. **810**	275. **50**
276. **2**	277. **1150**	278. **40**	279. **4**	280. **264**
281. **2**	282. **80**	283. **20**	284. **10**	285. **11**
286. **25**	287. **2**	288. **1**	289. **60**	290. **25**
291. **1**	292. **5**	293. **7**	294. **942**	295. **4**
296. **50**	297. **25**	298. **15**	299. **30**	300. **11**
301. **2**	302. **2**	303. **48**	304. **25**	305. **2**
306. **5**	307. **500**	308. **20**	309. **24**	310. **300**

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311. **25**	312. **6**	313. **363**	314. **100**	315. **8**
316. **500**	317. **40**	318. **2**	319. **6**	320. **20**
321. **3**	322. **32**	323. **25**	324. **2**	325. **25600**
326. **101**	327. **4**			

Out of syllabus

328. **A**	329. **B**	330. **C**	331. **D**	332. **C**
333. **D**	334. **B**	335. **B**	336. **D**	337. **C**
338. **A**	339. **D**	340. **28**	341. **16**	342. **57**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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